

ECE321

Electronic circuits

Lecture (05)- Op. Amp_introduction_Linear_Apps

Dr. Ahmed Reda^{1,2}

¹Department of Electronics and Communications, Faculty of Engineering, Zagazig University, Zagazig, Egypt.

²Ph.D, the MoE Key Laboratory of Artificial Intelligence & Bio-Circuits and Systems Laboratory (**BiCASL**), Department of Micro-Nano Electronics, Shanghai Jiao Tong University, Shanghai, China.



- ☐ Op Amp Definition
- ☐ Op Amp Characteristics
- ☐ Op Amp Models
- ☐ Applications of Op amp
 - ☐ Inverting Amplifier
 - ☐ Inverting Integrator Amplifier
 - ☐ Inverting Differentiator Amplifier
 - ☐ Non-Inverting Amplifier
 - ☐ Voltage Follower
 - ☐ Difference Amplifier
- ☐ Exercise

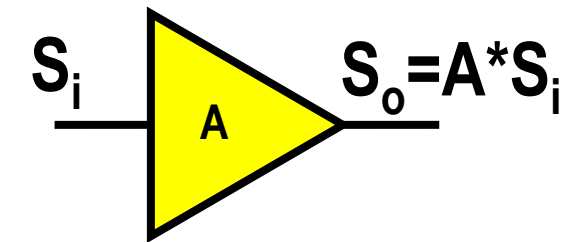


- ☐ **Op Amp Definition**
- ☐ **Op Amp Characteristics**
- ☐ **Op Amp Models**
- ☐ **Applications of Op amp**
 - ☐ **Inverting Amplifier**
 - ☐ **Inverting Integrator Amplifier**
 - ☐ **Inverting Differentiator Amplifier**
 - ☐ **Non-Inverting Amplifier**
 - ☐ **Voltage Follower**
 - ☐ **Difference Amplifier**
- ☐ **Exercise**

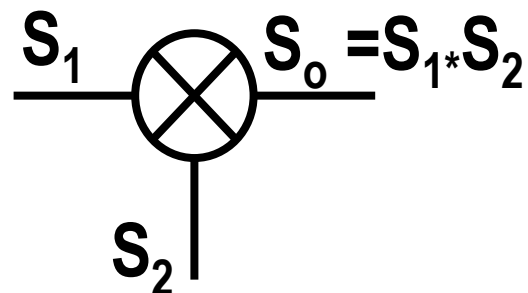
Op Amp Definition (1/4)



- ❑ An operational amplifier, or **op-amp**, is a very high gain differential amplifier with high input impedance and low output impedance.
- ❑ Op. Amp. is usually named as an **Universal building IP block** 😊
- ❑ **Typical uses** of the operational amplifier are to provide voltage amplitude changes (amplitude and polarity), **oscillators**, **filter** circuits, and many types of instrumentation circuits.



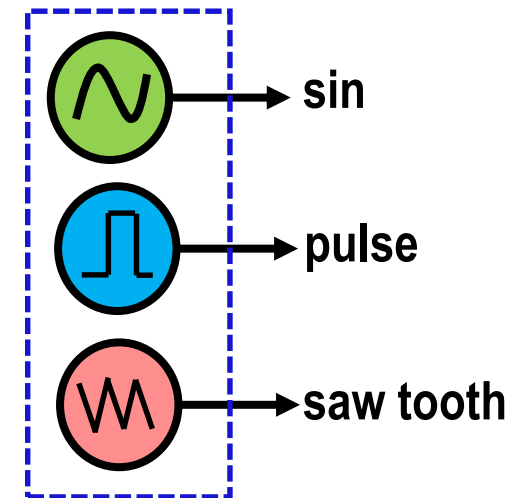
Amplify (V/I)



Mixer

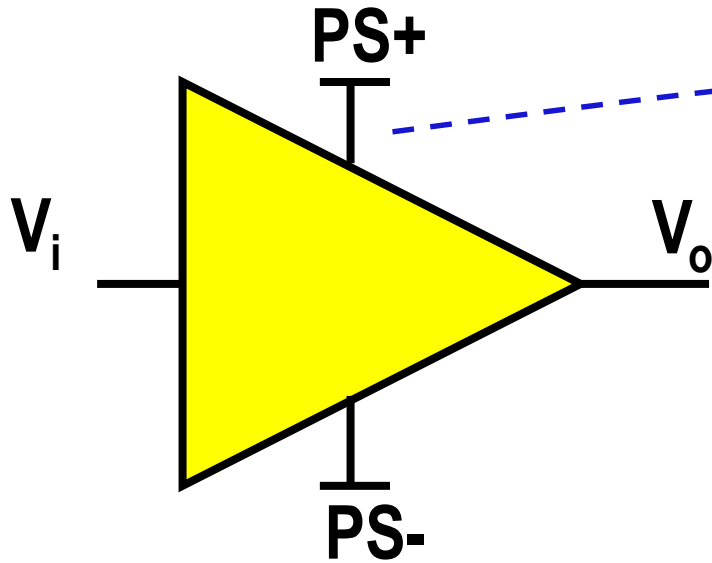
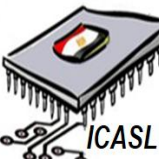


Filter



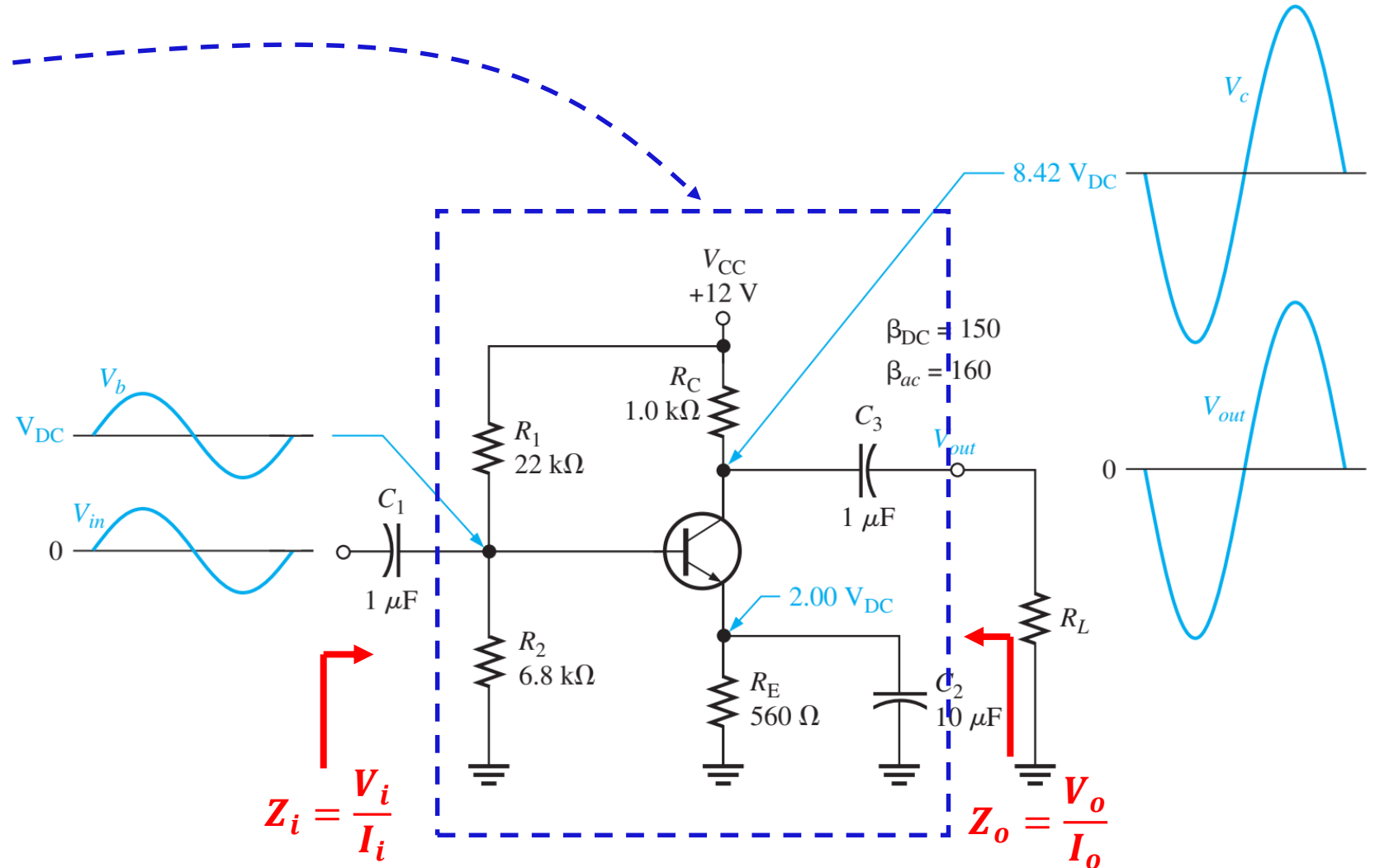
Oscillator/
Signal Wave Generator

Op Amp Definition (2/4)



AC Parameters:

Voltage Gain	$A_v = \frac{V_o}{V_i}$
Current Gain	$A_i = \frac{I_o}{I_i}$
Power Gain	$A_p = \frac{p_o}{p_i}$
Input Impedance	$Z_i = \frac{V_i}{I_i}$
Output Impedance	$Z_o = \frac{V_o}{I_o}$



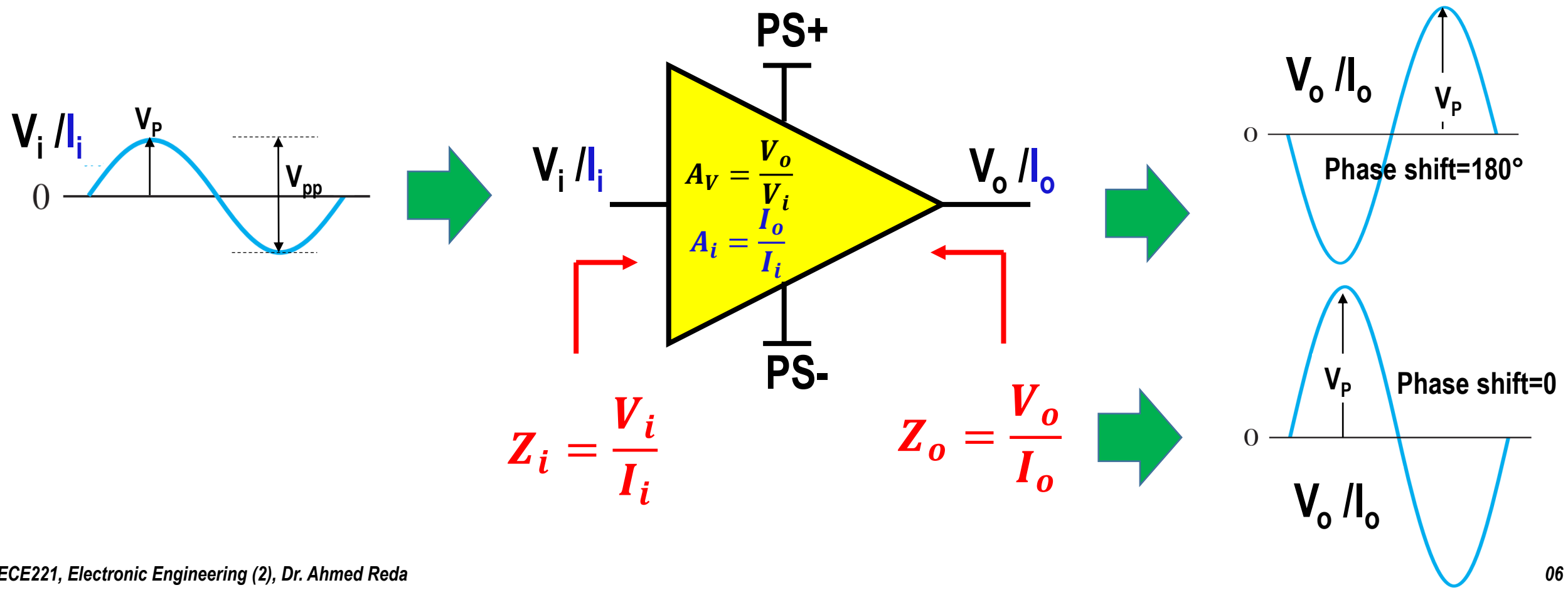
You should do DC analysis Firstly
Then, move to AC analysis



Op Amp Definition (3/4)



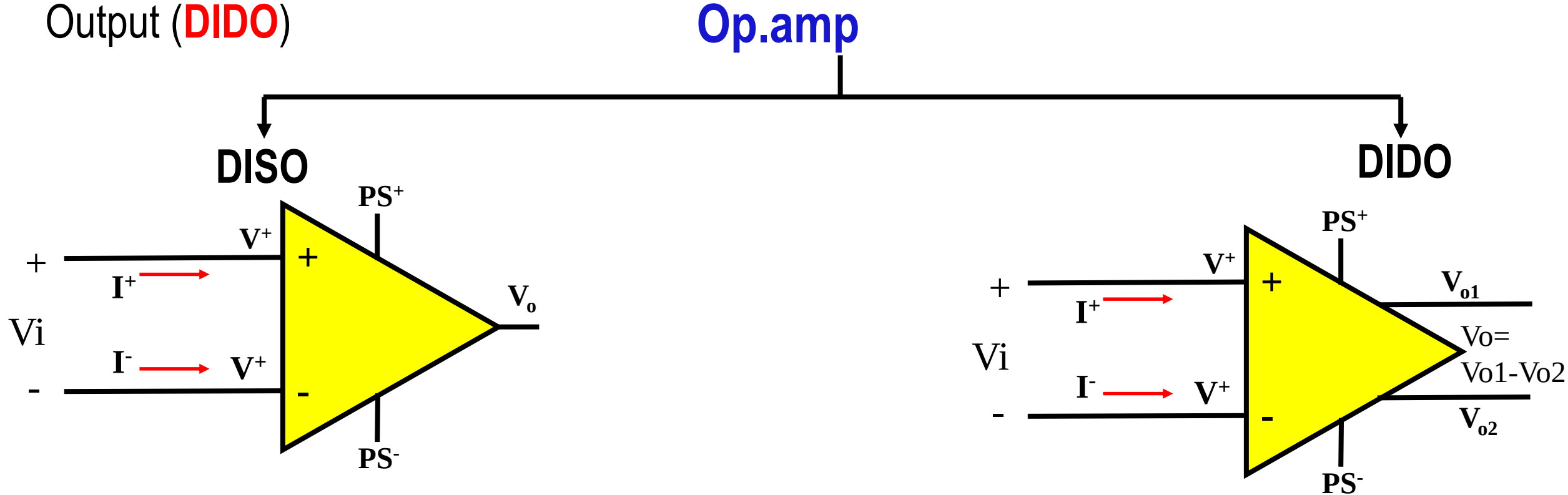
- ❑ The biasing of a transistor is purely **a dc operation**.
- ❑ The purpose of biasing is to establish **a Q-point in active region** to guarantee amplification action
- ❑ In applications where small signal voltages must be amplified—such as from an antenna or a microphone—variations about the Q-point are relatively small → *small-signal amplifiers*



Op Amp Definition (4/4)



- ❑ Op.amp could be Differential Input Single Output (**DISO**) or Differential Input Differential Output (**DIDO**)



V^+ : Non-inverting I/p voltage

V^- : Inverting I/P voltage

V_i : Differential input signal ($V^+ - V^-$)

PS^+ : Positive power supply (V_{CC}, V_{DD})

PS^- : Negative power supply (V_{EE}, V_{SS})

V_o : Output voltage

I^- : Inverting I/P current

I^+ : Non-inverting I/P current

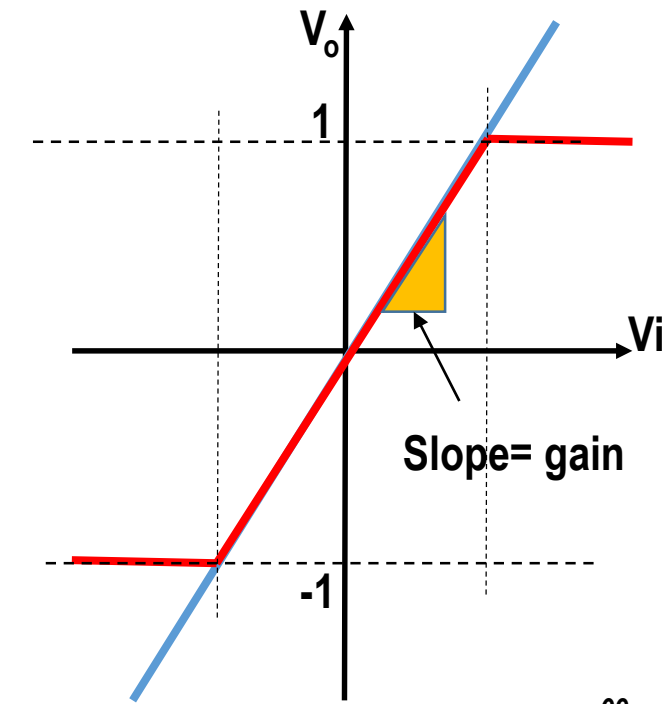
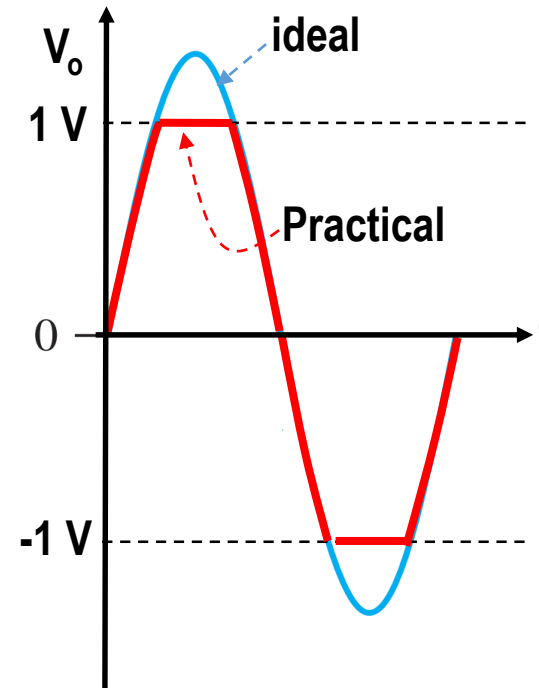
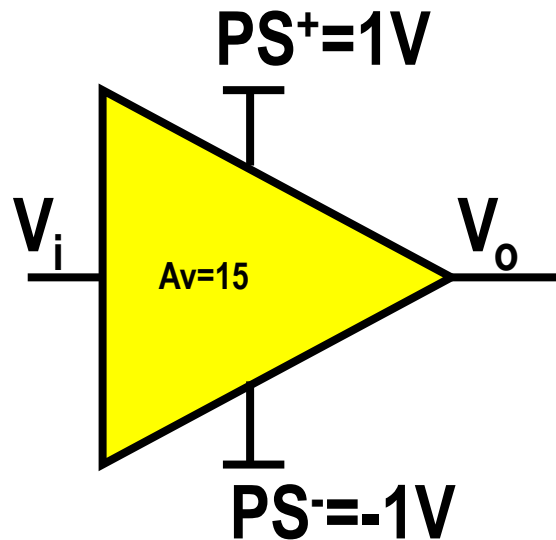
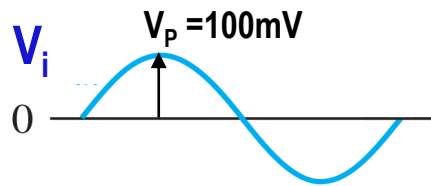


- ☐ Op Amp Definition
- ☐ **Op Amp Characteristics**
- ☐ Op Amp Models
- ☐ Applications of Op amp
 - ☐ Inverting Amplifier
 - ☐ Inverting Integrator Amplifier
 - ☐ Inverting Differentiator Amplifier
 - ☐ Non-Inverting Amplifier
 - ☐ Voltage Follower
 - ☐ Difference Amplifier
- ☐ Exercise

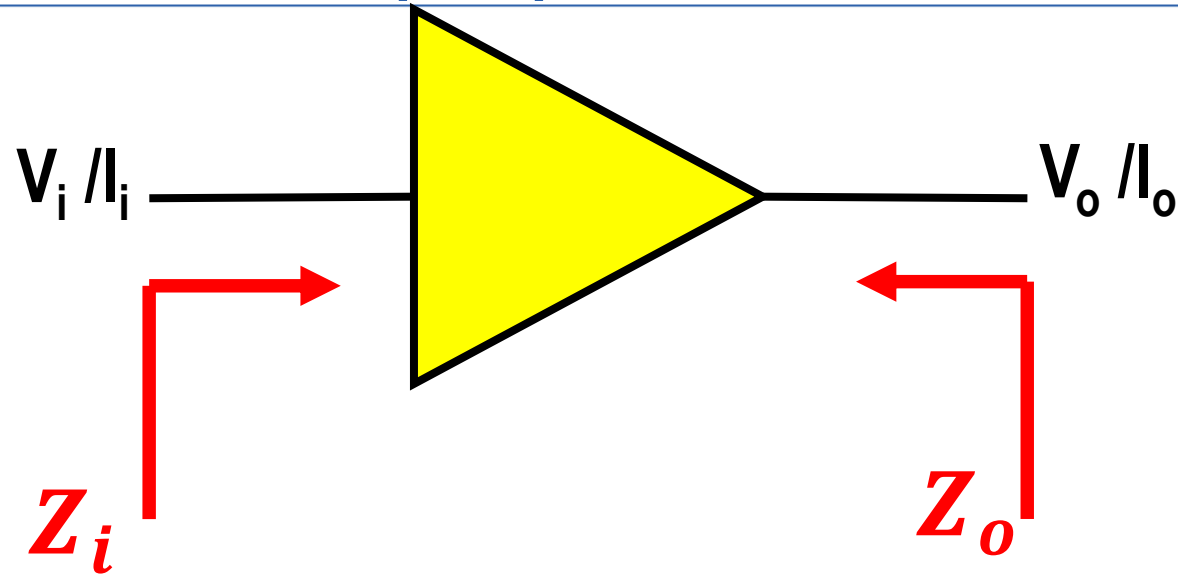
Op Amp Characteristics (1/2)



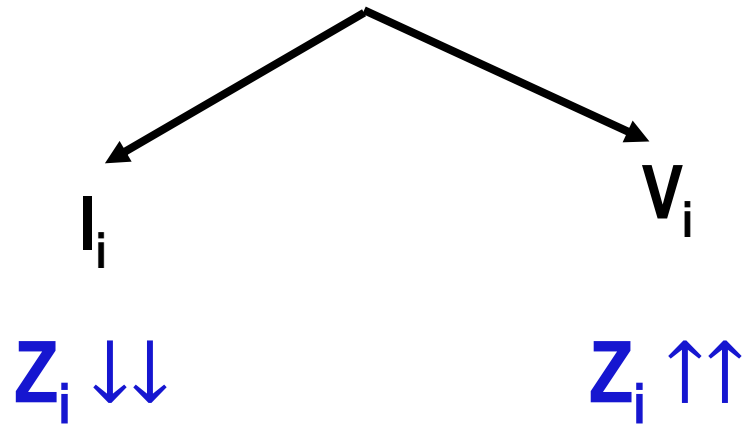
Ideal Ch/Cs	Practical Ch/Cs
$I^+ = 0$ $I^- = 0$ $\Rightarrow Z_{in} = \infty$ $V^+ = V^-$ $A_{v0} = \infty$ // open loop gain No distortion in the output signal (Linear) Bandwidth is infinite	$I^+ \neq 0$ $I^- \neq 0$ $\Rightarrow Z_{in} \neq \infty$ $V^+ \neq V^-$ $A_{v0} \neq \infty$ // open loop gain Output is limited to power supply Bandwidth is finite



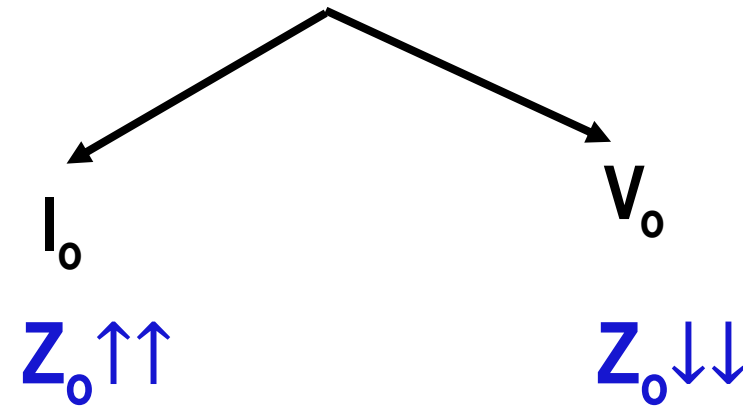
Op Amp Characteristics (2/2)



Op-amp can manipulate with input signal $\rightarrow V_i$ or I_i



Op-amp can generate output signal $\rightarrow V_o$ or I_o





- ☐ Op Amp Definition
- ☐ Op Amp Characteristics
- ☐ **Op Amp Models**
- ☐ Applications of Op amp
 - ☐ Inverting Amplifier
 - ☐ Inverting Integrator Amplifier
 - ☐ Inverting Differentiator Amplifier
 - ☐ Non-Inverting Amplifier
 - ☐ Voltage Follower
 - ☐ Difference Amplifier
- ☐ Exercise

Op Amp Models (1/2)



Type	Circuit Model	Ideal Ch/Cs
1-Voltage Amplifier $A_{vo} = \frac{V_o}{V_i}$		$Z_i \uparrow \uparrow \cong \infty$ $Z_o \downarrow \downarrow \cong 0$
2-Current Amplifier $A_{io} = \frac{I_o}{I_i}$		$Z_i \downarrow \downarrow \cong 0$ $Z_o \uparrow \uparrow \cong \infty$

Op Amp Models (2/2)

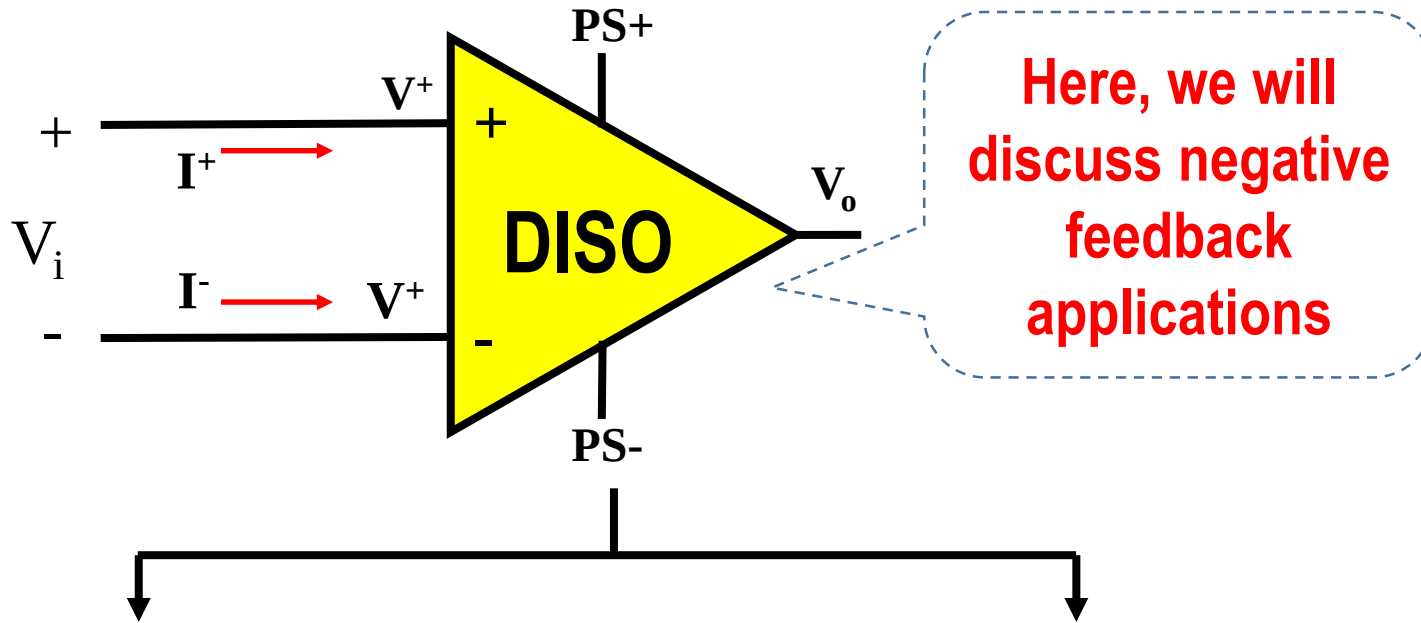


Type	Circuit Model	Ideal Ch/Cs
3- Trans-conductance Amplifier $G_m = \frac{I_o}{V_i}$		$Z_i \uparrow \uparrow \cong \infty$ $Z_o \uparrow \uparrow \cong \infty$
4- Trans- resistance Amplifier $R_m = \frac{V_o}{I_i}$		$Z_i \downarrow \downarrow \cong 0$ $Z_o \downarrow \downarrow \cong 0$



- ☐ Op Amp Definition
- ☐ Op Amp Characteristics
- ☐ Op Amp Models
- ☒ **Applications of Op amp**
 - ☐ Inverting Amplifier
 - ☐ Inverting Integrator Amplifier
 - ☐ Inverting Differentiator Amplifier
 - ☐ Non-Inverting Amplifier
 - ☐ Voltage Follower
 - ☐ Difference Amplifier
- ☐ Exercise

Applications of Op amp

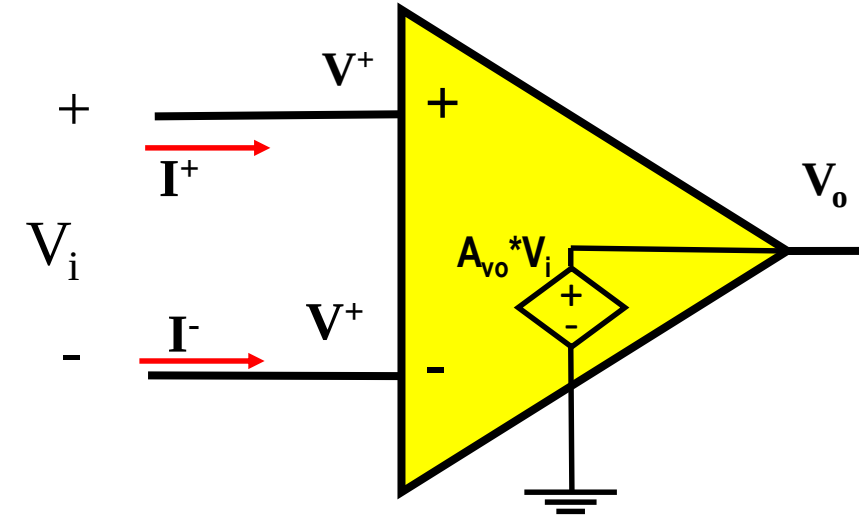


Linear Applications

- ☐ Inverting amplifier
- ☐ Inverting summing amplifier
- ☐ Inverting integrator amplifier
- ☐ Inverting differentiator amplifier
- ☐ Non-inverting amplifier
- ☐ Voltage follower amplifier
- ☐ Difference amplifier

Non-linear Applications

- ☐ Log Amplifier
- ☐ Anti-log Amplifier



For idea DISO amplifier

$$I^+ = I^- = 0$$

$$V^+ = V^-$$

$$V_o = A_{VO}(V^+ - V^-)$$

$$Z_i = \infty$$

$$Z_o = 0$$

$$A_{VO} = \infty$$



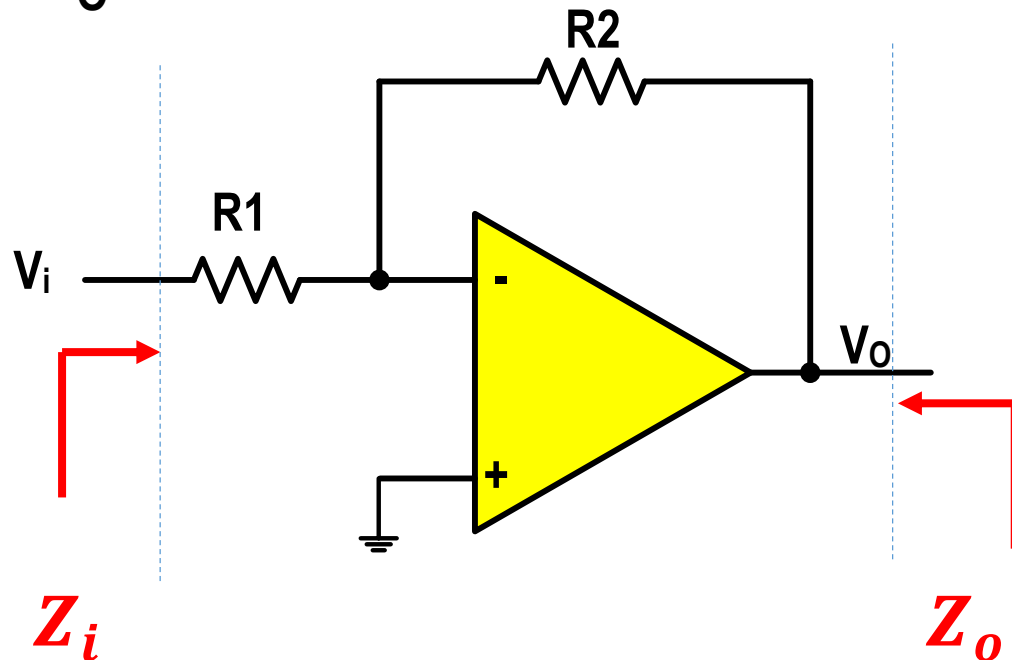
- ☐ Op Amp Definition
- ☐ Op Amp Characteristics
- ☐ Op Amp Models
- ☐ Applications of Op amp
 - ☐ **Inverting Amplifier**
 - ☐ Inverting Integrator Amplifier
 - ☐ Inverting Differentiator Amplifier
 - ☐ Non-Inverting Amplifier
 - ☐ Voltage Follower
 - ☐ Difference Amplifier
- ☐ Exercise

Inverting Amplifier (1/3)



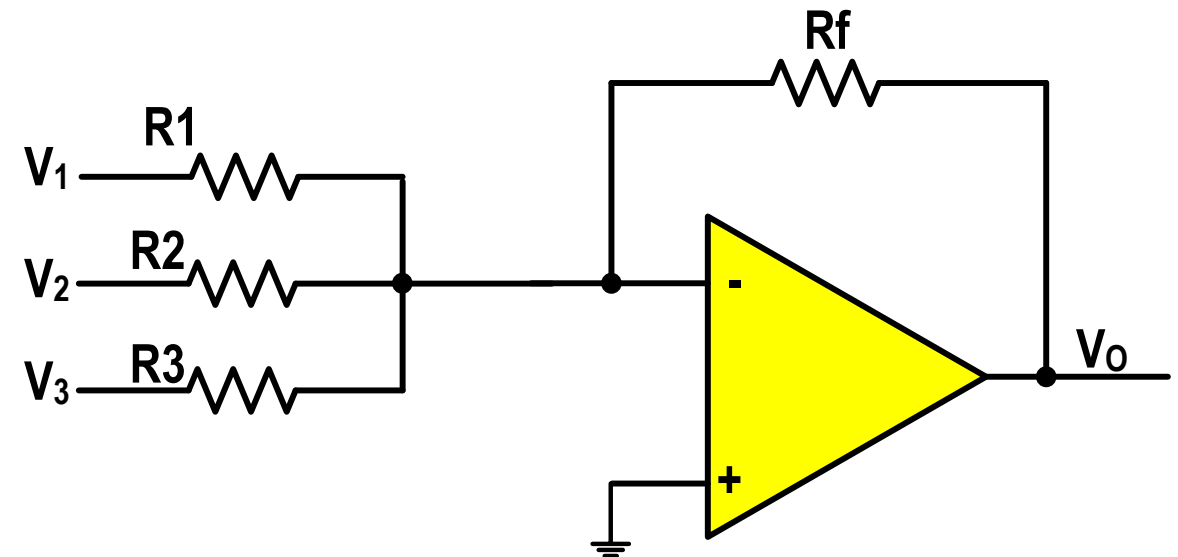
□ Find Expression

1. A_v "close loop gain"
2. Z_i
3. Z_o



Ans.

□ Find Expression of V_o for the summing circuit

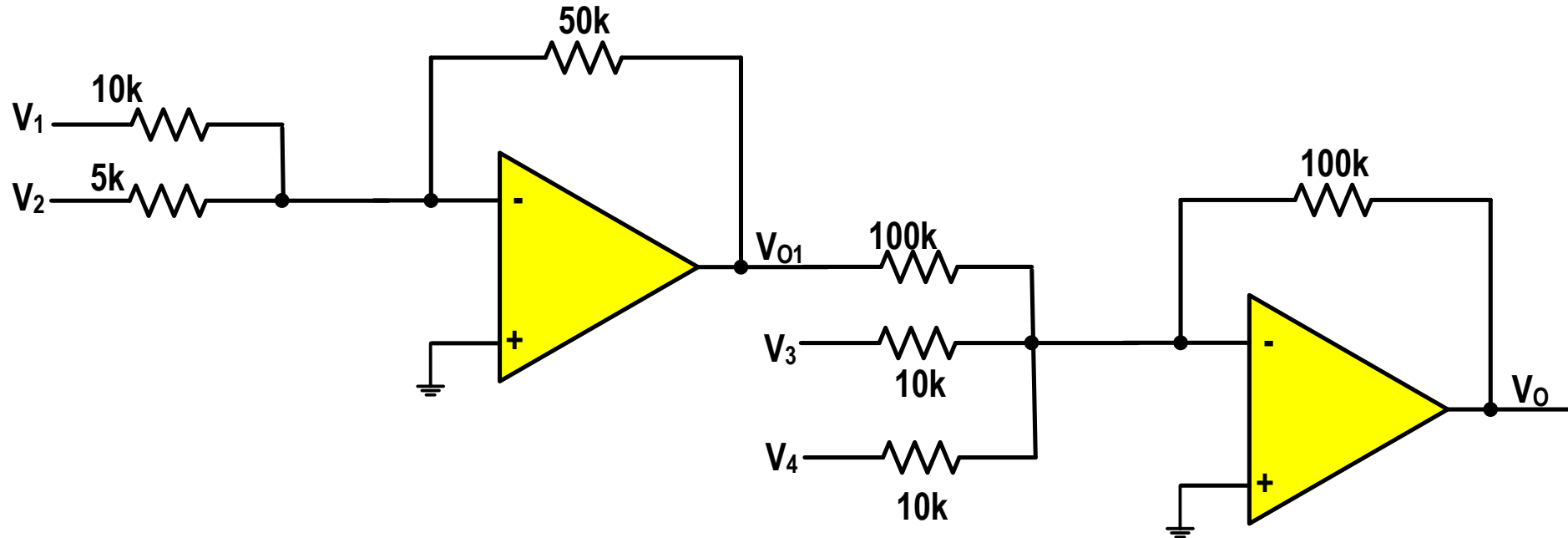


Ans.

Inverting Amplifier (2/3)



□ Find Expression of V_o for the next circuit



Ans.

$$V_o = 5V_1 + 10V_2 - 10V_3 - 10V_4$$



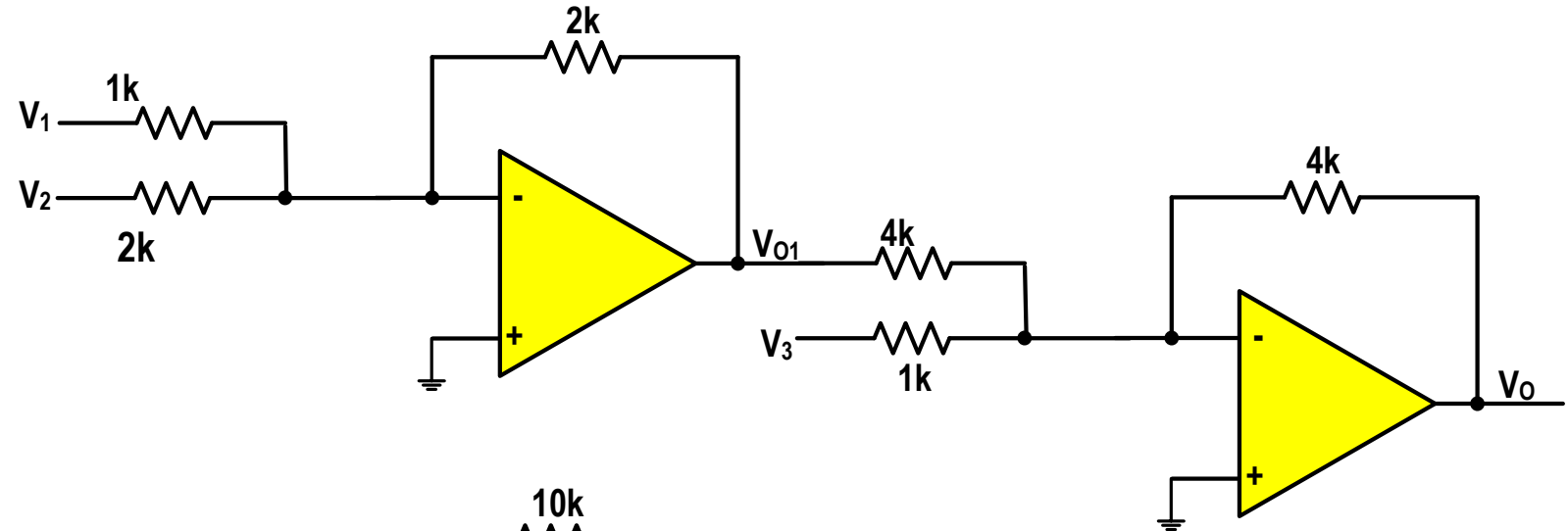
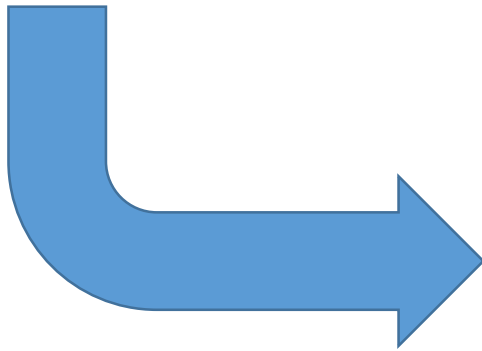
How

Inverting Amplifier (3/3)

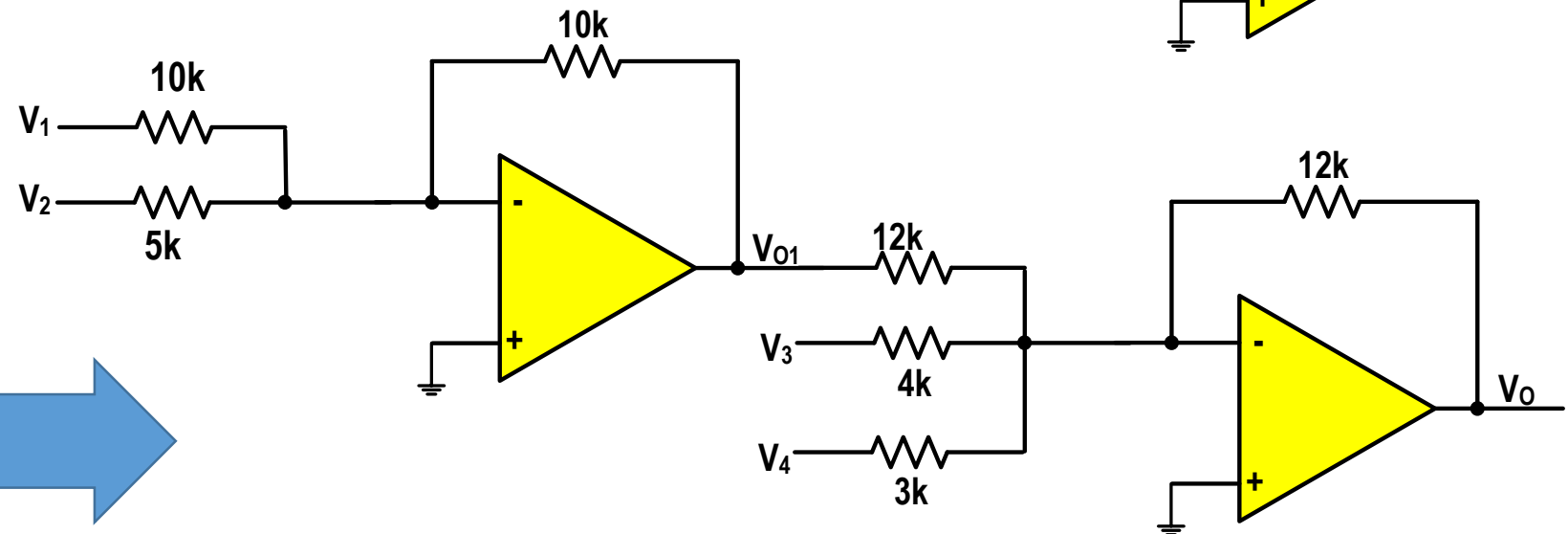
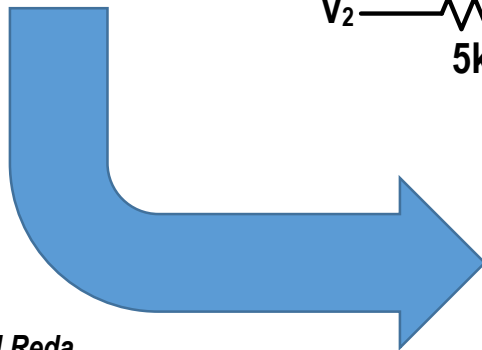


□ Using ideal op amps, try to implement the following function using inverting op-amp configuration.

1. $V_O = 2V_1 + V_2 - 4V_3$



2. $V_O = V_1 + 2V_2 - 3V_3 - 4V_4$





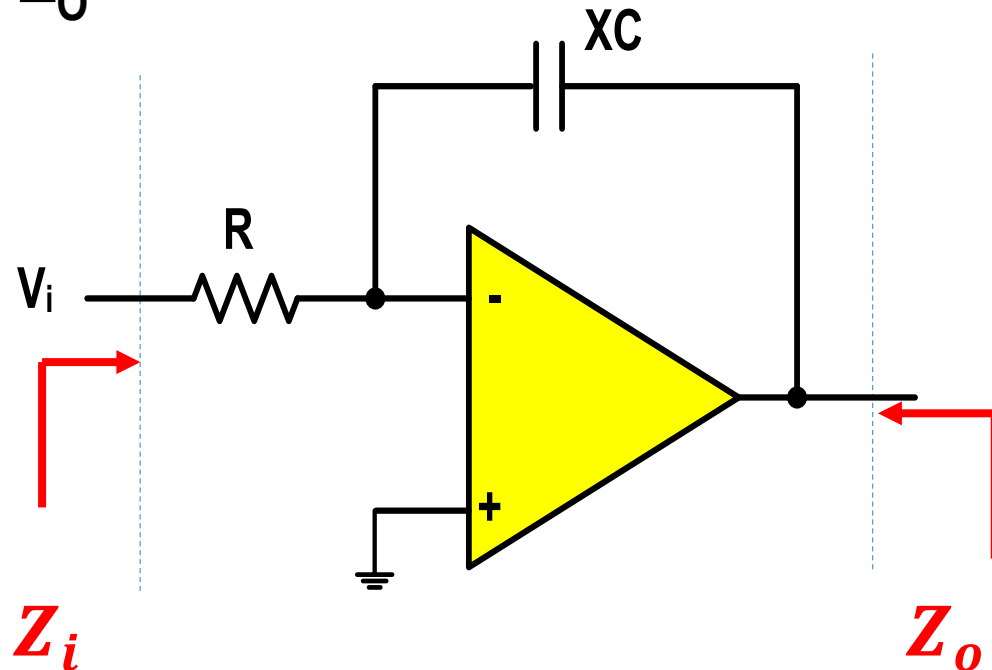
- ☐ Op Amp Definition
- ☐ Op Amp Characteristics
- ☐ Op Amp Models
- ☐ Applications of Op amp
 - ☐ Inverting Amplifier
 - ☐ Inverting Integrator Amplifier
 - ☐ Inverting Differentiator Amplifier
 - ☐ Non-Inverting Amplifier
 - ☐ Voltage Follower
 - ☐ Difference Amplifier
- ☐ Exercise

Inverting Integrator Amplifier

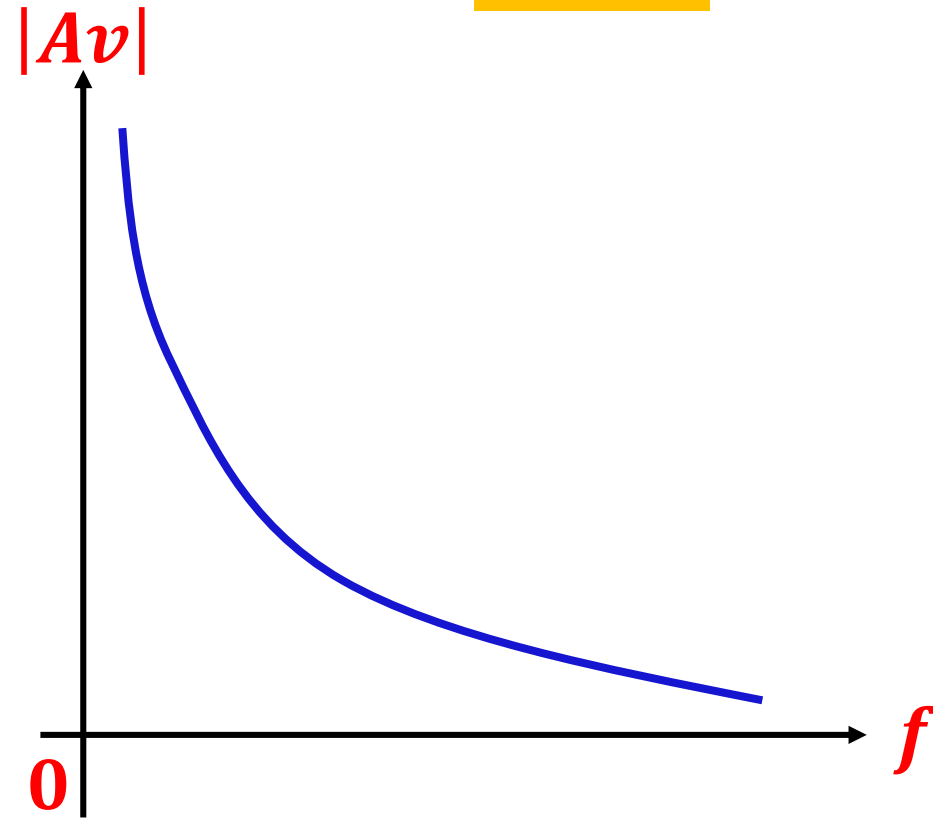


Find Expression

1. A_v "close loop gain"
2. Z_i
3. Z_o



Notes



- ❑ @DC, the integrator behaves $A_v \rightarrow \infty$
- ❑ To solve it, connect a resistor in parallel with the capacitor



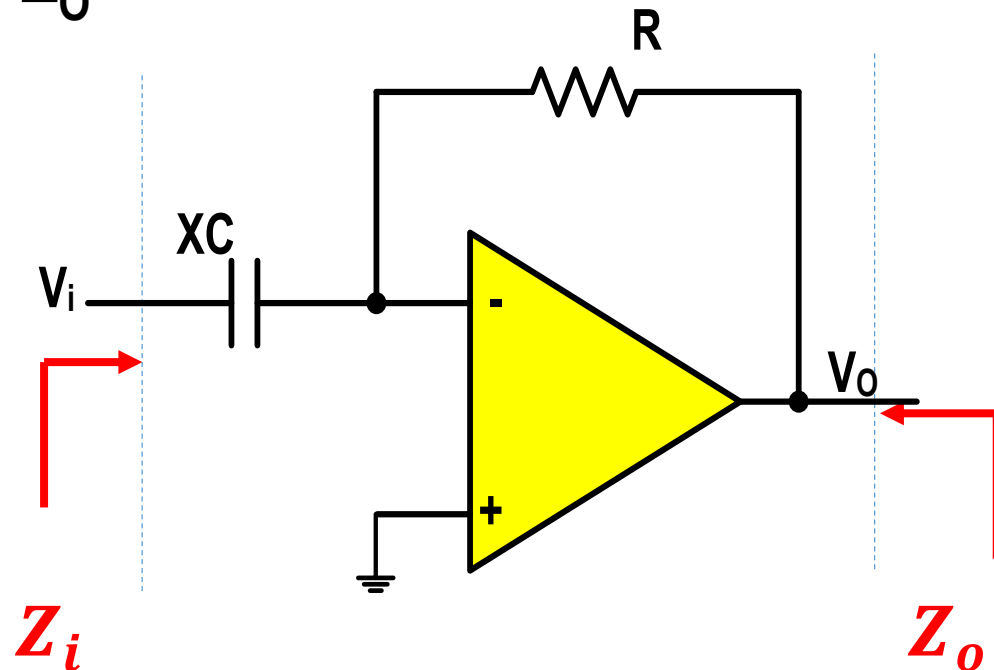
- ☐ Op Amp Definition
- ☐ Op Amp Characteristics
- ☐ Op Amp Models
- ☐ Applications of Op amp
 - ☐ Inverting Amplifier
 - ☐ Inverting Integrator Amplifier
 - ☐ Inverting Differentiator Amplifier
 - ☐ Non-Inverting Amplifier
 - ☐ Voltage Follower
 - ☐ Difference Amplifier
- ☐ Exercise

Inverting Differentiator Amplifier

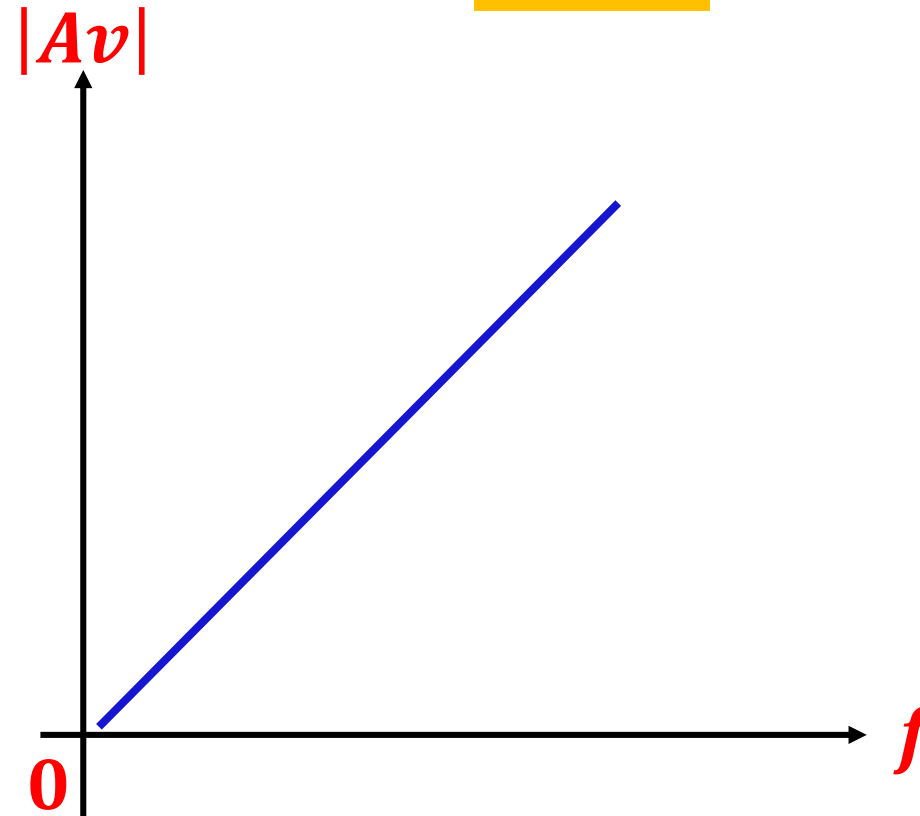


□ Find Expression

1. A_v “close loop gain”
2. Z_i
3. Z_o



Notes



- @DC, the differentiator behaves $A_v \rightarrow 0$
- To solve it, connect a resistor in parallel with the capacitor



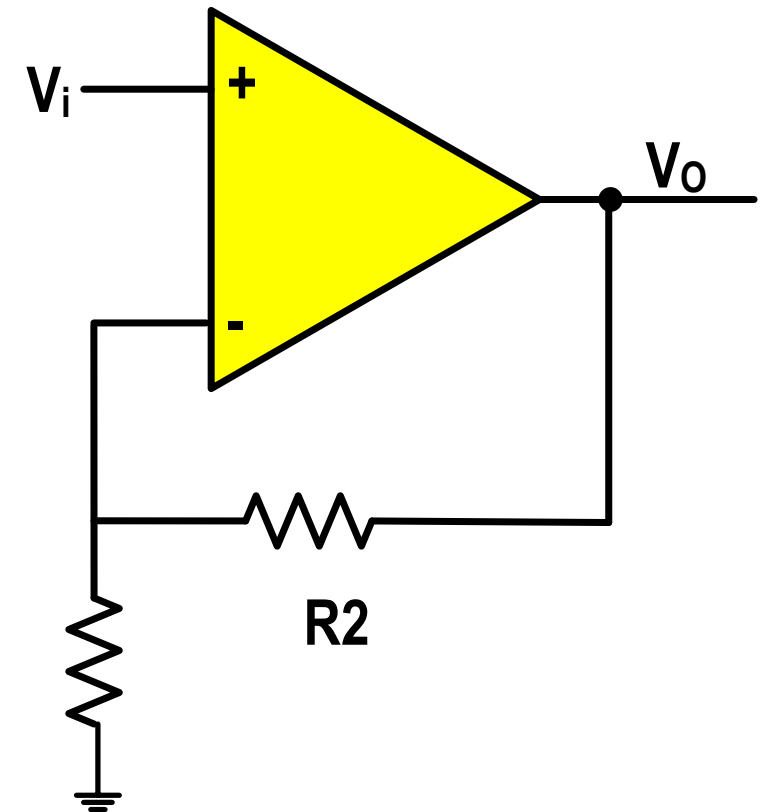
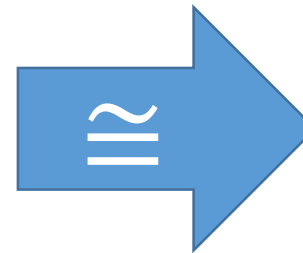
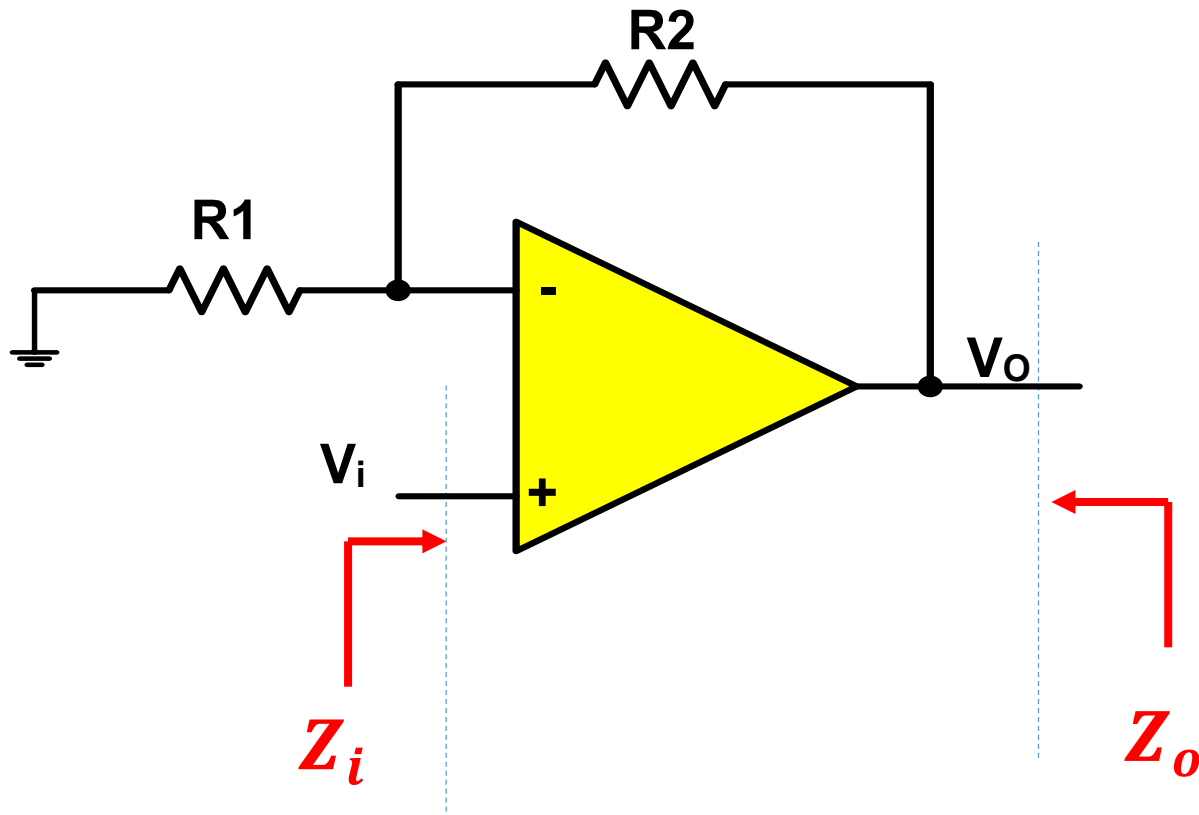
- ☐ Op Amp Definition
- ☐ Op Amp Characteristics
- ☐ Op Amp Models
- ☐ Applications of Op amp
 - ☐ Inverting Amplifier
 - ☐ Inverting Integrator Amplifier
 - ☐ Inverting Differentiator Amplifier
 - ☐ Non-Inverting Amplifier
 - ☐ Voltage Follower
 - ☐ Difference Amplifier
- ☐ Exercise

Non-Inverting Amplifier



□ Find Expression

1. A_v "close loop gain"
2. Z_i
3. Z_o





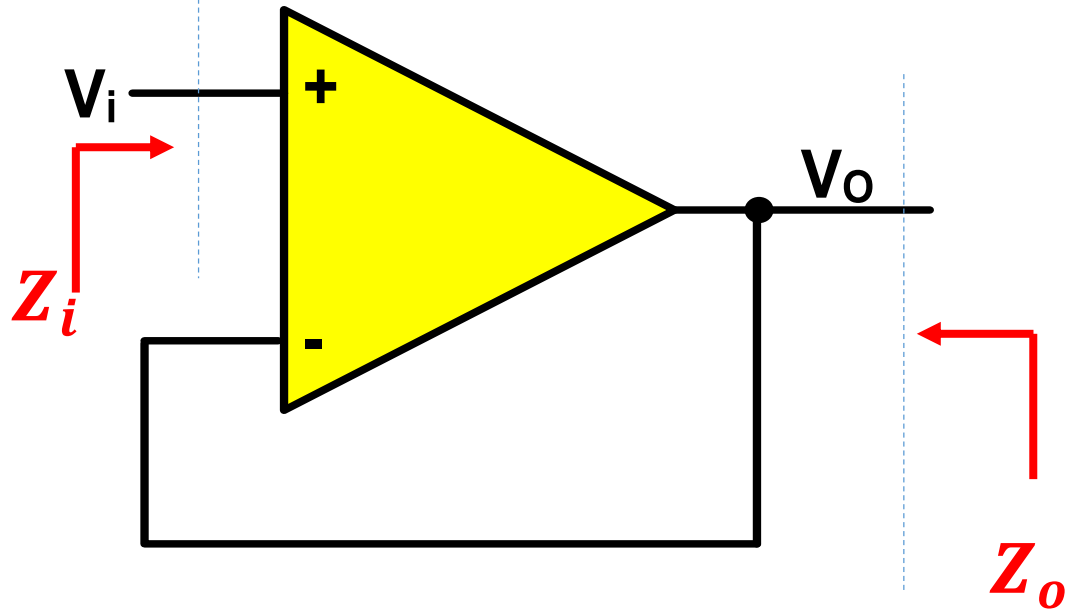
- ☐ Op Amp Definition
- ☐ Op Amp Characteristics
- ☐ Op Amp Models
- ☐ Applications of Op amp
 - ☐ Inverting Amplifier
 - ☐ Inverting Integrator Amplifier
 - ☐ Inverting Differentiator Amplifier
 - ☐ Non-Inverting Amplifier
 - ☐ **Voltage Follower**
 - ☐ Difference Amplifier
- ☐ Exercise

Voltage Follower



□ Find Expression

1. A_v “close loop gain”
2. Z_i
3. Z_o





- ☐ Op Amp Definition
- ☐ Op Amp Characteristics
- ☐ Op Amp Models
- ☐ Applications of Op amp
 - ☐ Inverting Amplifier
 - ☐ Inverting Integrator Amplifier
 - ☐ Inverting Differentiator Amplifier
 - ☐ Non-Inverting Amplifier
 - ☐ Voltage Follower
 - ☐ Difference Amplifier
- ☐ Exercise

Difference Amplifier (1/5)

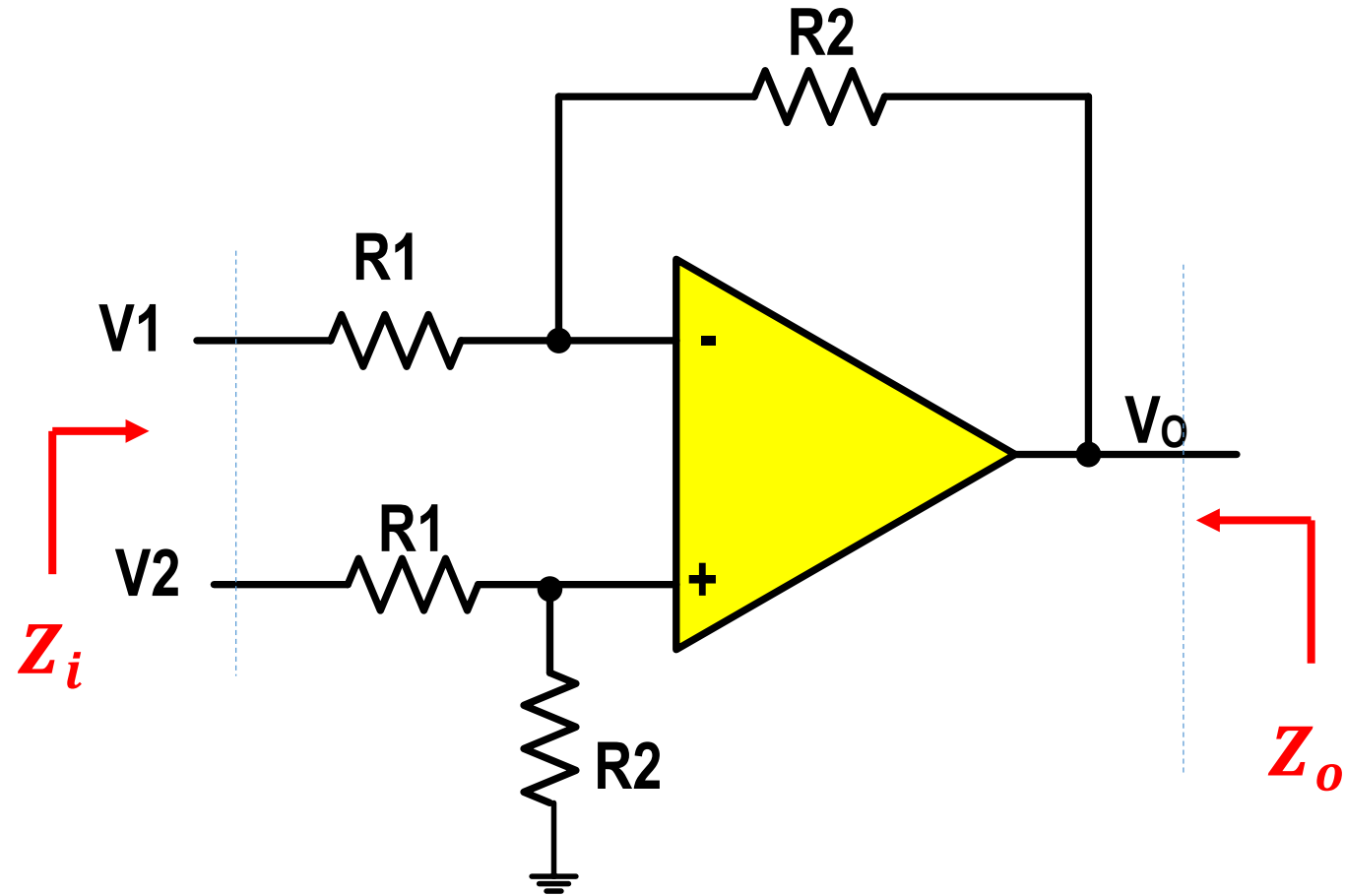


□ Derive Expression for

1. $V_o = \frac{R_2}{R_1} (V_2 - V_1)$

2. $Z_i = 2 R_1$

3. $Z_o = 0$

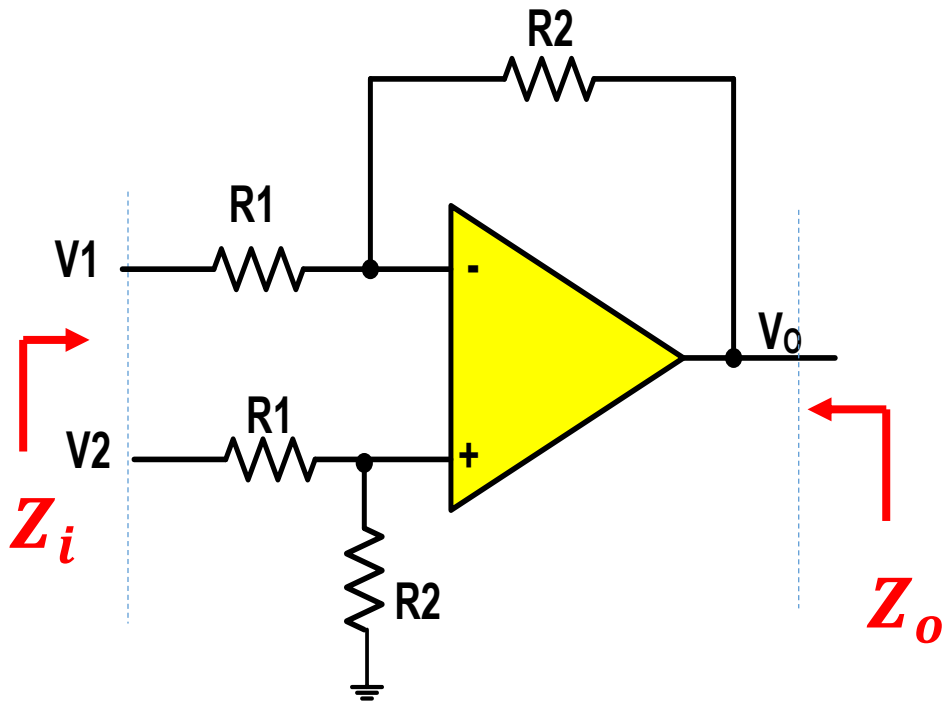


Difference Amplifier (2/5)



□ Derive Expression for

1. $V_o = \frac{R_2}{R_1} (V_2 - V_1)$
2. $Z_i = 2 R_1$
3. $Z_o = 0$



Ans.

For linear application, we can use super position theory

$$V_o = V_{o1} |_{\text{due to } V_1} + V_{o2} |_{\text{due to } V_2}$$

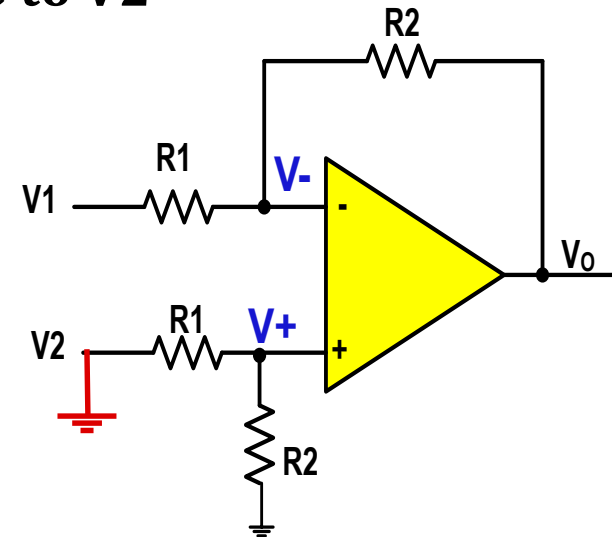
□ To get V_{o1}

For ideal op amp

$$I^+ = I^- = 0$$

$$V^+ = V^- = 0$$

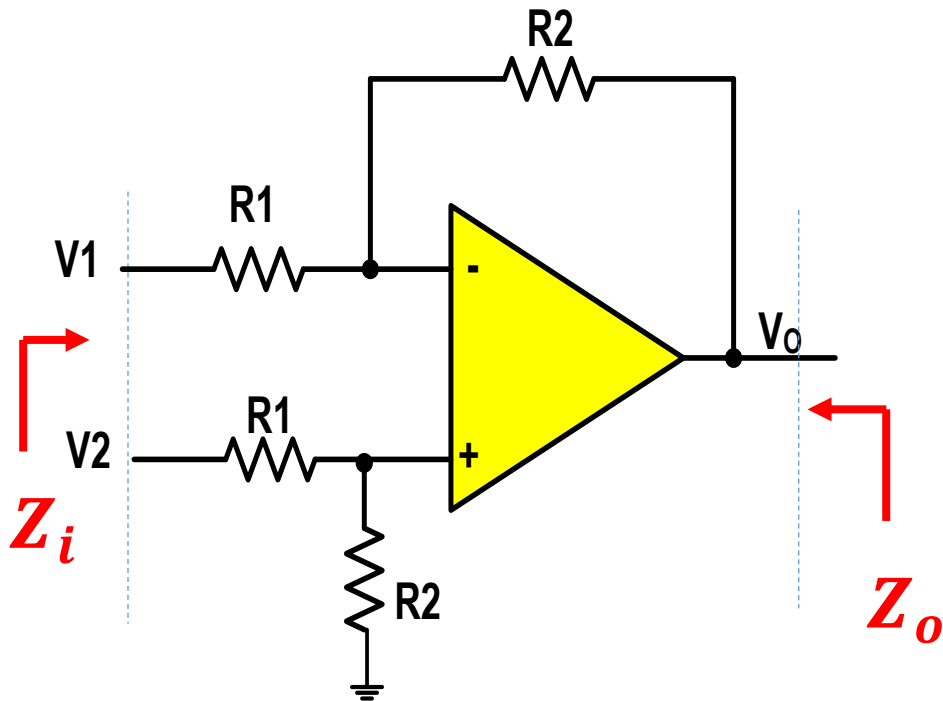
$$V_{o1} = \frac{-R_2}{R_1} V_1 \longrightarrow \textcircled{1}$$



Difference Amplifier (3/5)

□ Derive Expression for

1. $V_o = \frac{R_2}{R_1} (V_2 - V_1)$
2. $Z_i = 2 R_1$
3. $Z_o = 0$



Ans.

□ To get V_{o2}

For ideal op amp

$$I^+ = I^- = 0$$

$$V^+ = V^- = \frac{R_2}{R_2 + R_1} V_2$$

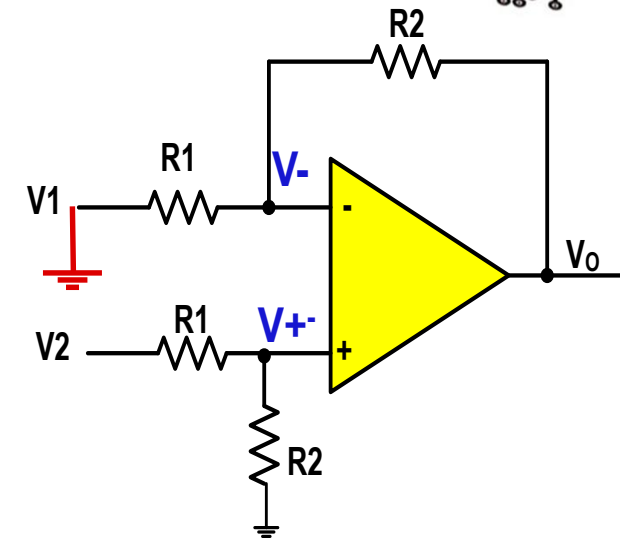
$$V_{o2} = V^+ \left(1 + \frac{R_2}{R_1}\right)$$

$$V_{o2} = \frac{R_2}{R_2 + R_1} V_2 \left(1 + \frac{R_2}{R_1}\right)$$

$$V_{o2} = \frac{R_2}{R_2 + R_1} V_2 \left(\frac{R_1 + R_2}{R_1}\right)$$

$$V_{o2} = \frac{R_2}{R_1} V_2 \longrightarrow \textcircled{2}$$

From (1) and (2)



$$V_o = \frac{-R_2}{R_1} V_1 + \frac{R_2}{R_1} V_2$$



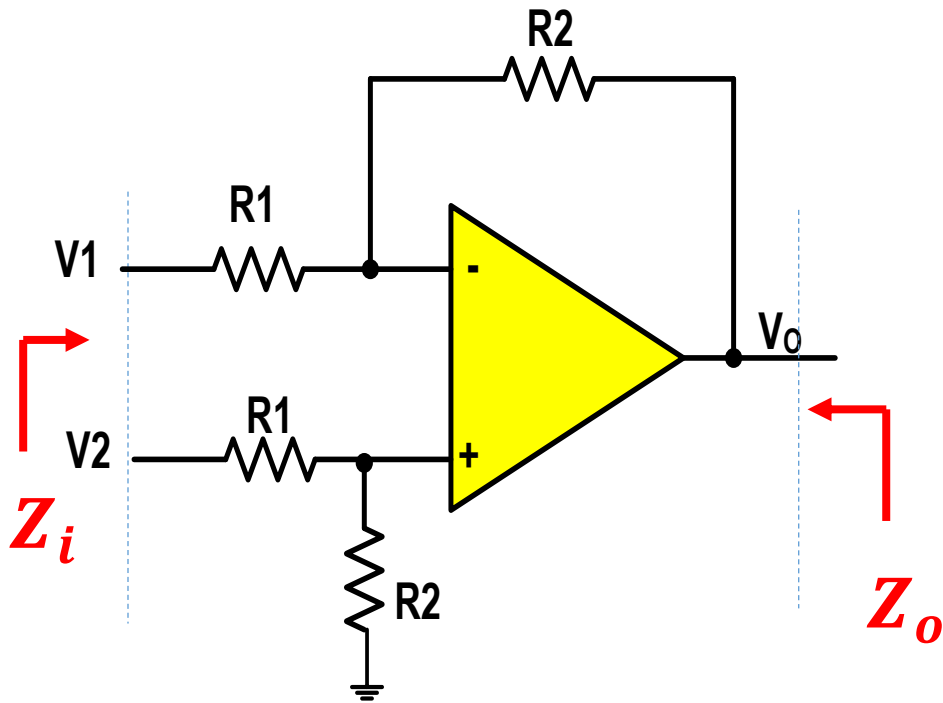
$$V_o = \frac{R_2}{R_1} (V_2 - V_1)$$

Difference amplifier

Difference Amplifier (4/5)

□ Derive Expression for

1. $V_o = \frac{R_2}{R_1} (V_2 - V_1)$
2. $Z_i = 2 R_1$
3. $Z_o = 0$



Ans.

□ To get Z_i

$$Z_i = \frac{V_{id}}{I_i}$$

For loop(1)

$$V_1 - I_1 * R_1 - I_2 * R_1 - V_2 = 0$$

$$V_1 - V_2 = I_1 * R_1 + I_2 * R_1$$

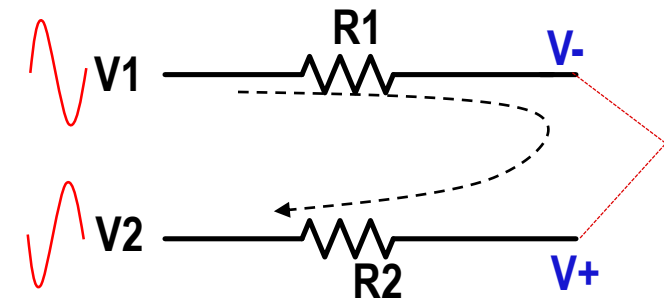
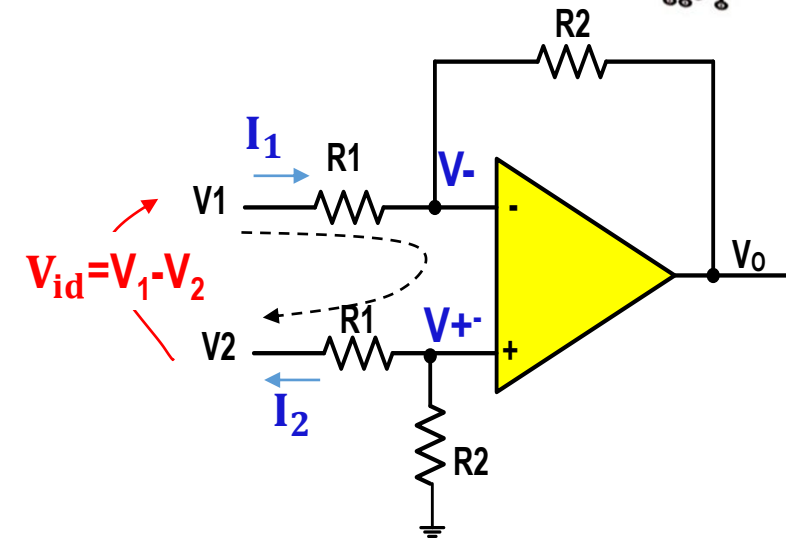
$$V_{id} = R_1 (I_1 + I_2)$$

$$\text{Where, } I_1 = I_2 = I_i$$

$$V_{id} = 2 R_1 I_i$$



$$Z_i = \frac{V_{id}}{I_i} = 2 R_1$$

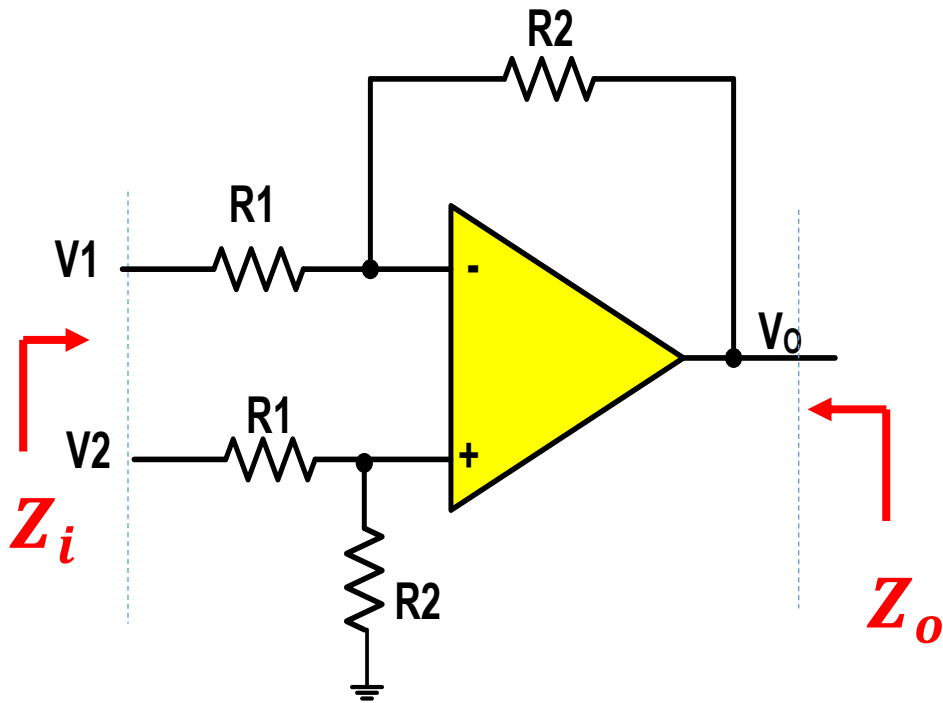


Difference Amplifier (5/5)



□ Derive Expression for

1. $V_o = \frac{R_2}{R_1} (V_2 - V_1)$
2. $Z_i = 2 R_1$
3. $Z_o = 0$



Ans.

□ To get Z_o

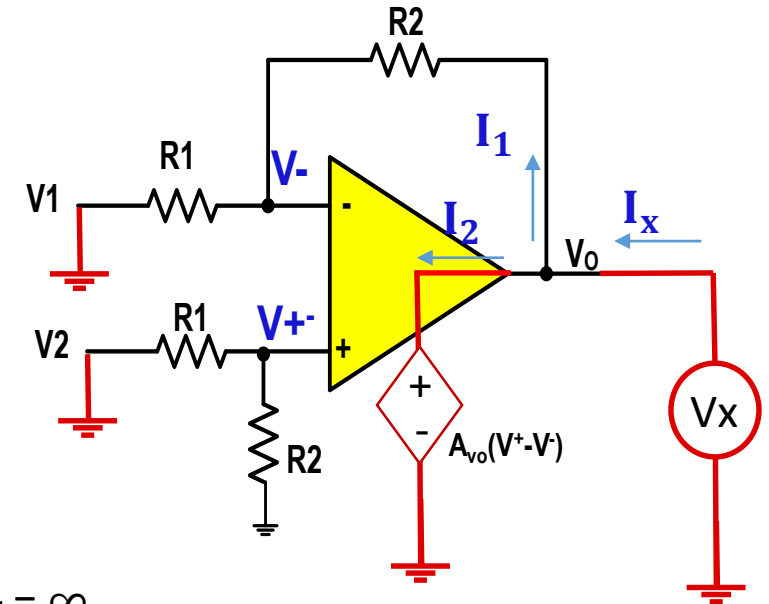
$$Z_o = \frac{V_x}{I_x}$$

Take node at V_x

$$I_x = I_1 + I_2$$

$$I_x = \frac{V_x - V^-}{R_2} + \frac{V_x - A_{vo}(V^+ - V^-)}{0} = \infty$$

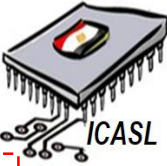
$$Z_o = \frac{V_x}{\infty} = 0$$





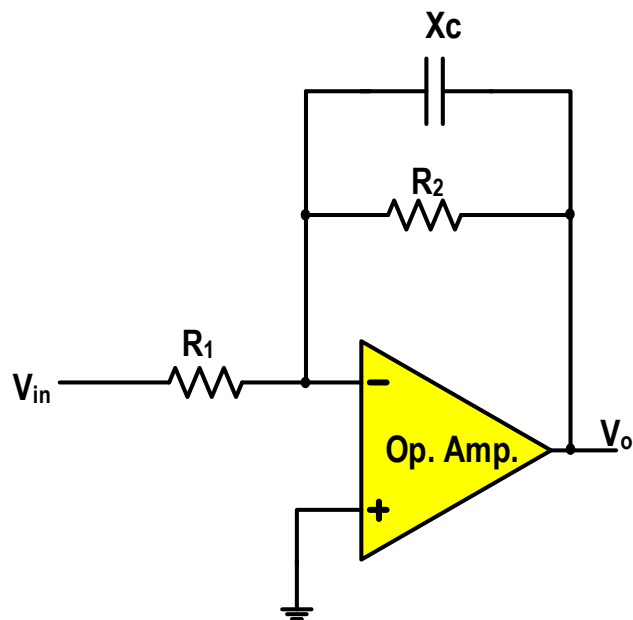
- ☐ Op Amp Definition
- ☐ Op Amp Characteristics
- ☐ Op Amp Models
- ☐ Applications of Op amp
 - ☐ Inverting Amplifier
 - ☐ Inverting Integrator Amplifier
 - ☐ Inverting Differentiator Amplifier
 - ☐ Non-Inverting Amplifier
 - ☐ Voltage Follower
 - ☐ Difference Amplifier
- ☐ Exercise

Exercise (1/6)



□ Ex.1, Derive Expression for

1. $H(s)=A_v$, $H(j\omega)=A_v$,and $|H(j\omega)| = |A_v|$
2. DC-gain
3. Cut-off frequency
4. Sketch the freq. Response
5. Sketch the phase Response



Ans.

□ To get $H(s)$, $H(j\omega)$,and $|H(j\omega)|$

For ideal op amp

$$V^+ = V^- = 0$$

$$V^+ = V^- = 0$$

$$H(s) = \frac{V_o}{V_{in}} = \frac{-Z}{R_1}$$

$$= \frac{X_c || R_2}{R_1} = \frac{-1}{R_1} * \frac{R_2 * X_c}{R_2 + X_c}$$

$$= \frac{-1}{R_1} * \frac{R_2 * \frac{1}{sC}}{R_2 + \frac{1}{sC}} = \frac{-R_2}{R_1} * \frac{1}{sCR_2 + 1}$$

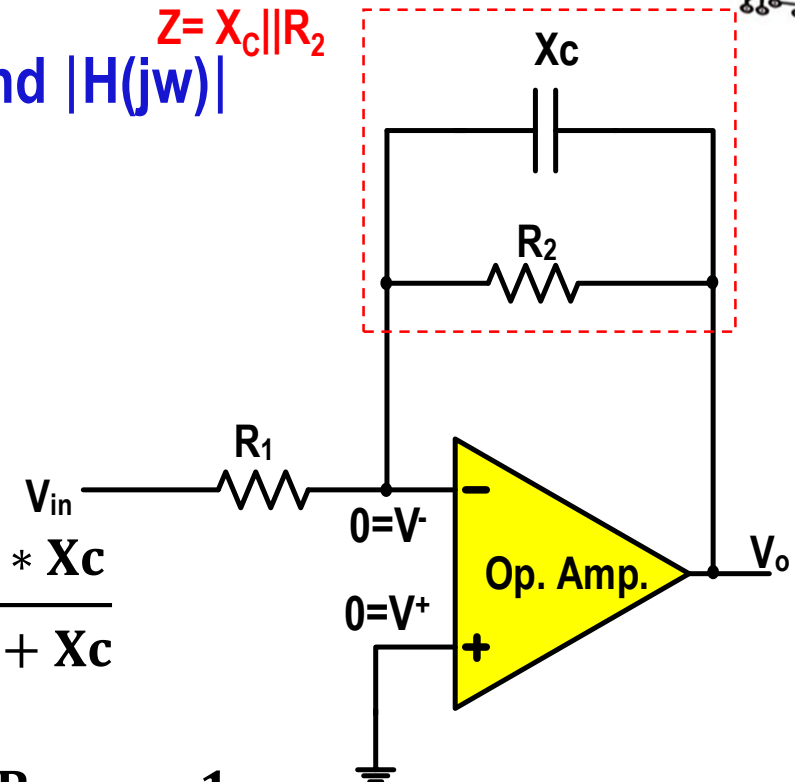
$$H(s) = \frac{-R_2}{R_1} * \frac{1}{sCR_2 + 1}$$

$$H(j\omega) = \frac{-R_2}{R_1} * \frac{1}{j\omega CR_2 + 1}$$

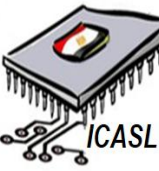
$$|H(j\omega)| = \frac{R_2}{R_1} * \frac{1}{\sqrt{1 + (\omega CR_2)^2}}$$

$$|H(j\omega)| = \frac{R_2}{R_1} * \frac{1}{\sqrt{1 + (\frac{\omega}{\omega_c})^2}}$$

$$Z = X_c || R_2$$



Exercise (2/6)



□ To DC-gain

$$|H(jw)| = \frac{R_2}{R_1} * \frac{1}{\sqrt{1 + (\frac{w}{wc})^2}}$$

$$|H(jw)| = \frac{R_2}{R_1} * \frac{1}{\sqrt{1 + (\frac{f}{fc})^2}}$$

$$\text{Dc_Gain} = \frac{R_2}{R_1}$$

@ w=0 rad/s ,f=0 Hz

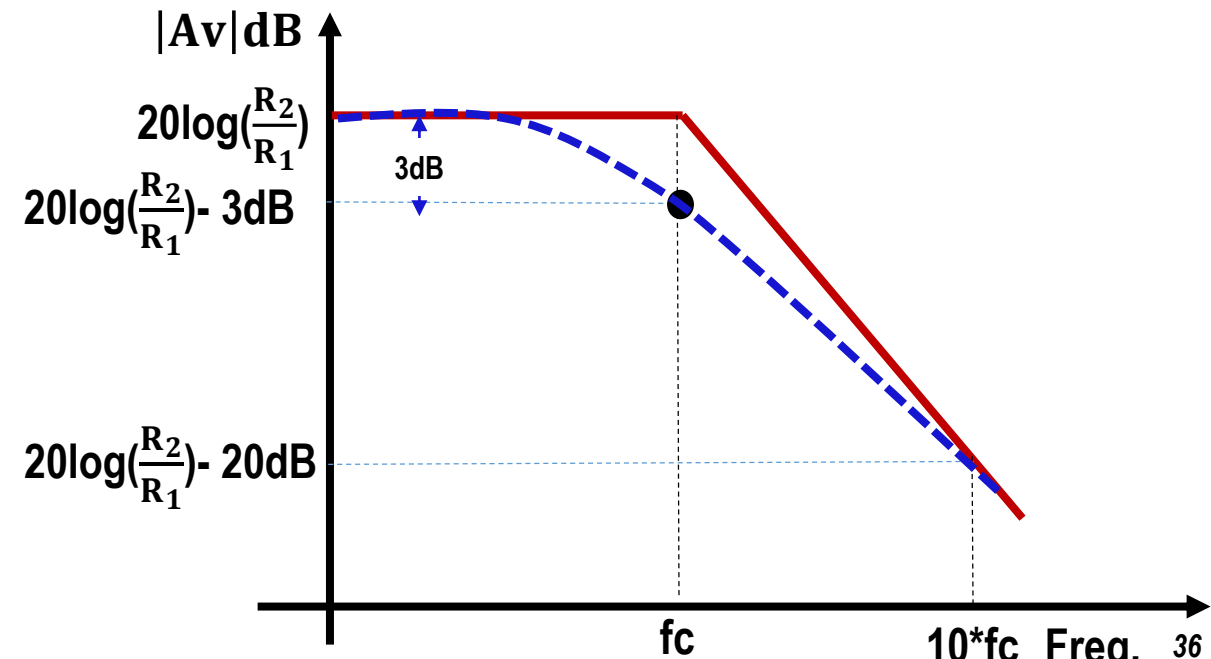
□ To Cut-off frequency

$$wc = \frac{1}{CR_2} \quad \longrightarrow \quad fc = \frac{1}{2\pi CR_2}$$

□ To Sketch the freq. Response

$$|Av| = \frac{R_2}{R_1} * \frac{1}{\sqrt{1 + (\frac{f}{fc})^2}}$$

f	Av	Av dB
f=0	$\frac{R_2}{R_1}$	$20\log(\frac{R_2}{R_1})$
f=0.1fc	$0.995 * \frac{R_2}{R_1}$	$\sim 20\log(\frac{R_2}{R_1})$
f=fc	$\frac{R_2}{\sqrt{2} R_1}$	$20\log(\frac{R_2}{R_1}) - 3\text{dB}$
f=10fc	$0.099 * \frac{R_2}{R_1}$	$20\log(\frac{R_2}{R_1}) - 20\text{dB}$
F=∞	0	-∞



Exercise (3/6)



□ To Sketch the phase Response

$$\phi(A_v) = \phi_{\text{numerator}} - \phi_{\text{denominator}}$$

$$= \tan^{-1}\left(\frac{\text{imag. part}}{\text{Real part}}\right) - \tan^{-1}\left(\frac{\text{imag. part}}{\text{Real part}}\right)$$

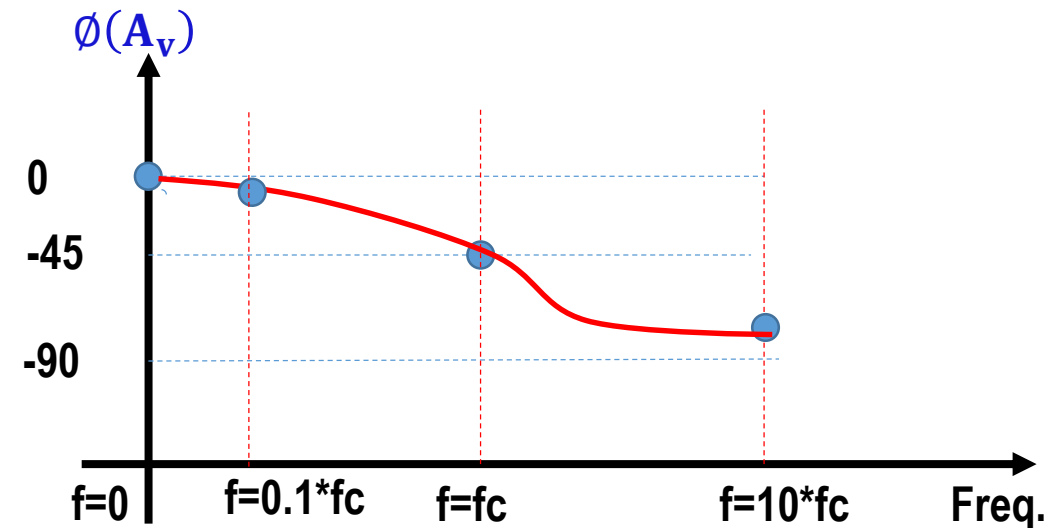
$$A_v = \frac{-R_2}{R_1} * \frac{1}{j\omega CR_2 + 1}$$

$$= \tan^{-1}\left(\frac{0}{-R_2}\right) - \tan^{-1}\left(\frac{\omega CR_2}{1}\right)$$

$$\phi(A_v) = -\tan^{-1}\left(\frac{\omega}{\omega_c}\right)$$

$$\phi(A_v) = -\tan^{-1}\left(\frac{f}{f_c}\right)$$

f	$\phi(A_v)$
0	0
0.1* f _c	-5.7
f _c	-45
10* f _c	-84
∞	-90



Exercise (3/6) Correction



❑ To Sketch the phase Response

$$\phi(A_v) = \phi_{\text{numerator}} - \phi_{\text{denominator}}$$

$$= \tan^{-1}\left(\frac{\text{imag. part}}{\text{Real part}}\right) - \tan^{-1}\left(\frac{\text{imag. part}}{|\text{Real part}|}\right)$$

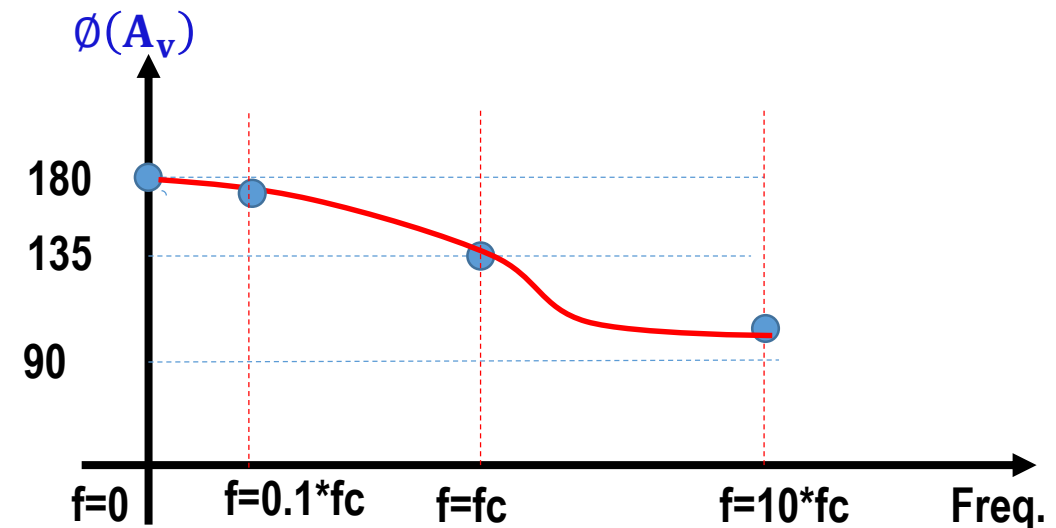
$$A_v = -\frac{\frac{R_2}{R_1}}{j\omega CR_2 + 1}$$

$$= 180 - \tan^{-1}\left(\frac{0}{\frac{R_2}{R_1}}\right) - \tan^{-1}\left(\frac{\omega CR_2}{1}\right)$$

$$\phi(A_v) = 180 - \tan^{-1}\left(\frac{\omega}{\omega_c}\right)$$

$$\phi(A_v) = 180 - \tan^{-1}\left(\frac{f}{f_c}\right)$$

f	$\phi(A_v)$
0	180
0.1* f _c	174.3
f _c	135
10* f _c	96
∞	90

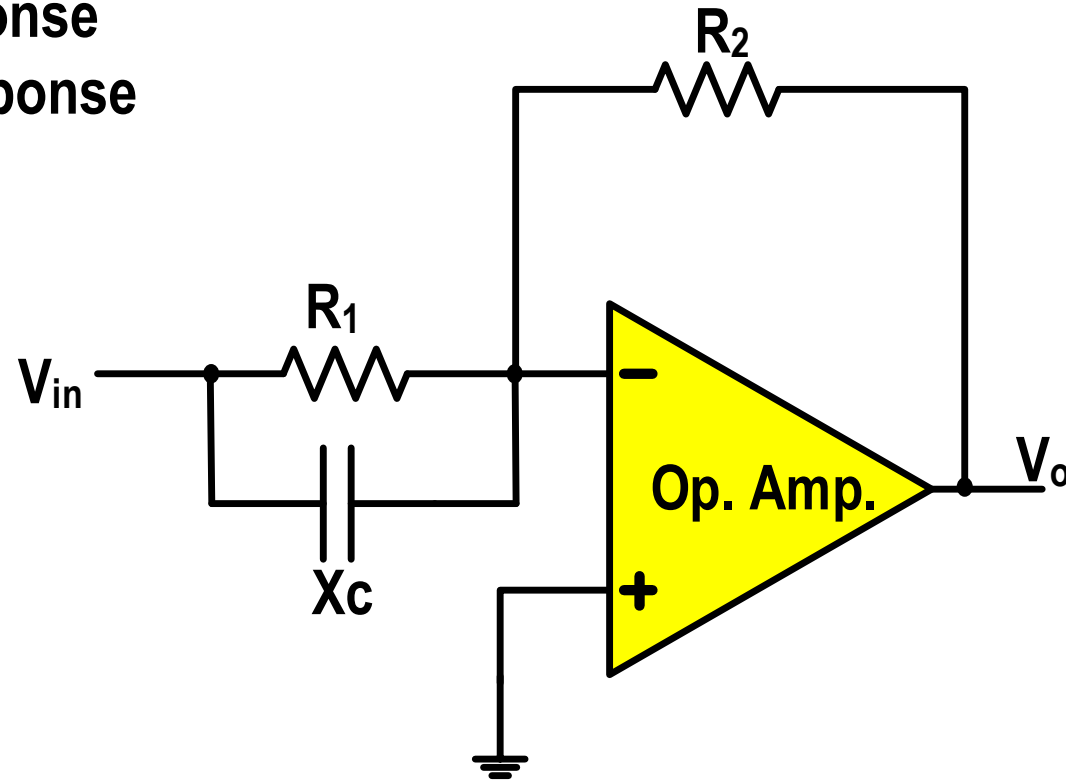


Exercise (4/6)



□ Ex2. Derive Expression for

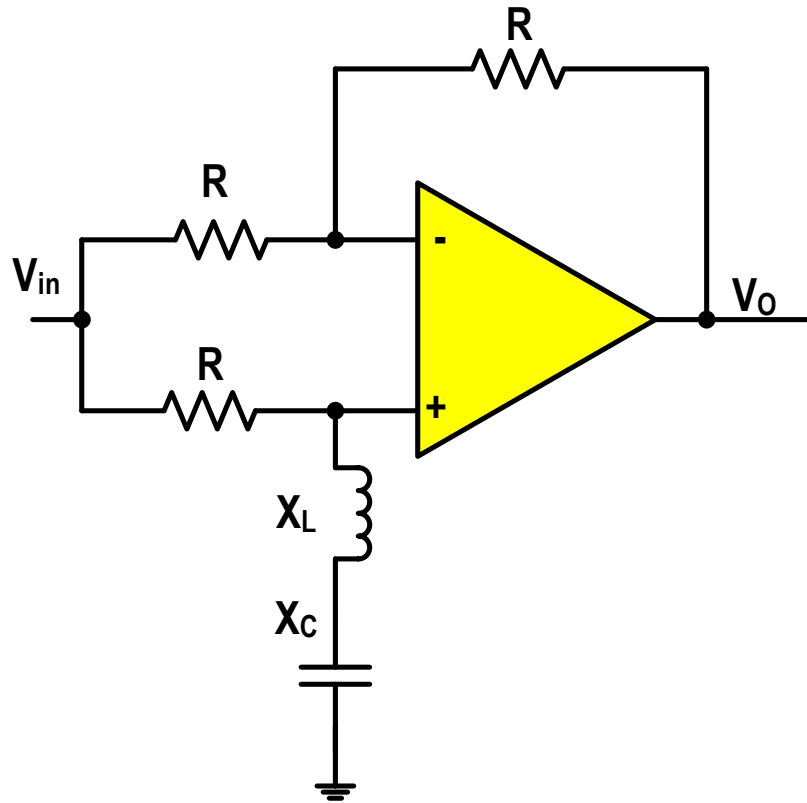
1. $H(s)=A_v$, $H(j\omega)=A_v$,and $|H(j\omega)| = |A_v|$
2. High frequency -gain
3. Cut-off frequency
4. Sketch the freq. Response
5. Sketch the phase Response



Exercise (5/6)

□ Ex.3, Derive Expression for

1. $H(s)=Av$, $H(jw)=Av$
2. Dc-gain



Ans.

For ideal op amp

$$I^+ = I^- = 0$$

$$V^+ = V^- = V_x$$

Take node (1)

$$I_1 + I_2 + I^-$$

$$\frac{v_x - v_{in}}{R} + \frac{v_x - v_o}{R} + 0 = 0$$

$$v_x \left(\frac{1}{R} + \frac{1}{R} \right) = \frac{v_{in}}{R} + \frac{v_o}{R}$$

$$v_x \left(\frac{2}{R} \right) = \frac{v_{in}}{R} + \frac{v_o}{R}$$

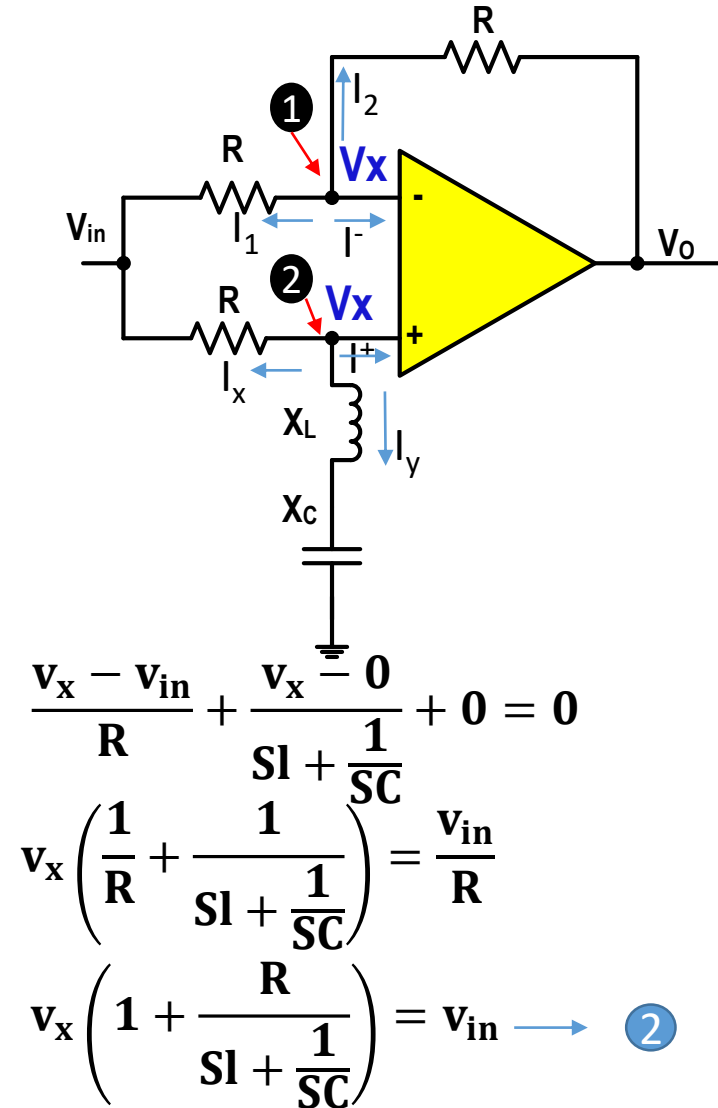
$$2 * v_x = v_{in} + v_o$$

$$v_x = 0.5 (v_{in} + v_o) \rightarrow \textcircled{1}$$

Take node (2)

$$I_x + I_y + I^+$$

$$= 0$$

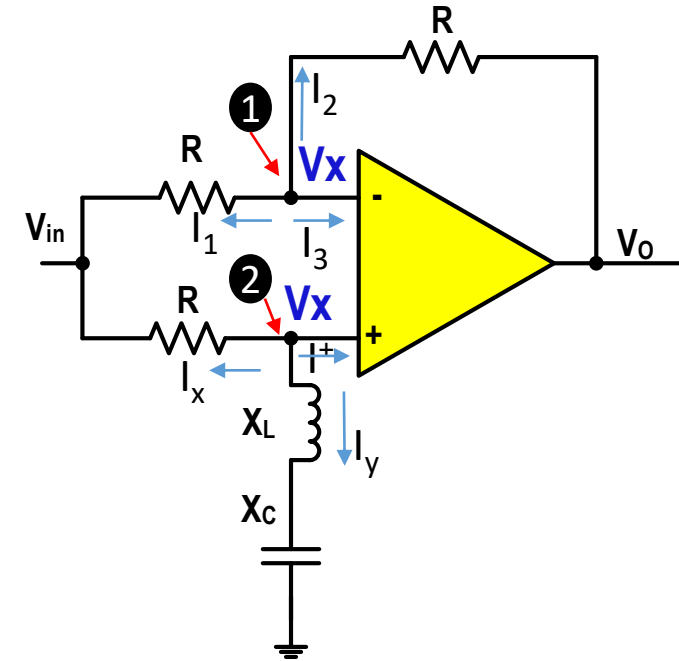


From (1) and (2)

1. $H(s)=Av$, $H(jw)=Av$

ECE321, Electronic circuits, Dr. Ahmed Reda

$$H(j\omega) = A_v = -1 + \left(\frac{2}{1 + \frac{R}{j\omega L + \frac{1}{j\omega SC}}} \right)$$

$$A_v = 1$$


Thanks!